

TR400

Industrial Router

User Manual

TesproOS 1.0000

Document Version 1.0

Tespro Electronics Co., Ltd.

www.tespro.com

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Chapter 1 Product Overview

The TR400 series is an industrial-grade router professionally developed by Tespro Electronics Co., Ltd. in line with the Industry 4.0 trend, designed to support enterprise digital transformation. It integrates Wi-Fi, cellular networks (4G/5G), rich IoT interfaces, VPN private network technology, and more, providing stable, secure, efficient, and intelligent network communication solutions for various IoT applications. Equipped with the self-developed TesproOS operating system designed specifically for IoT connectivity, the TR400 combines high-performance hardware with an intuitive Web management interface for convenient configuration and monitoring, enabling comprehensive control over devices, networks, protocols, and remote access.

Whether deployed in industrial facilities, retail stores, or remote offices, the TR400 delivers stable and reliable network connectivity through multiple WAN access methods including Gigabit Ethernet and 4G/5G cellular networks. It also supports serial port communication, Modbus protocol gateway, BACnet gateway, and energy management functions, making it ideal for IoT and industrial automation applications.

Tip: This version (v1.0) introduces VPN connection management, BACnet gateway service, and energy management service, with improvements to other chapters.

1.1 Key Features

- Dual-core ARM Cortex-A53 @ 1.3GHz with hardware NAT acceleration
- 512MB memory + 256MB storage
- WiFi 6 AX3000 dual-band: 2.4GHz 574Mbps + 5GHz 2402Mbps
- 5 Gigabit Ethernet ports (4 LAN + 1 WAN)
- Built-in cellular expansion interface for 4G/5G connectivity
- 3 serial ports (2× RS485 + 1× RS232) with UDP multicast service support
- Modbus RTU/TCP gateway supporting both Master and Slave roles; northbound channels support MQTT/TCP/UDP/HTTP/OPC UA Client/OPC UA Server
- BACnet gateway service supporting BACnet/IP and BACnet MS/TP southbound data collection
- Energy management service with real-time multi-device energy consumption monitoring and alerting
- VPN connection management supporting OpenVPN, WireGuard, IPsec, and PPTP
- USB 3.0 / USB 2.0 interfaces for storage and cellular expansion
- Industrial-grade firewall with zone-based security policies
- Application-layer traffic identification (NetCtrl/DPI engine)
- Modern Web management interface with dark/light theme switching

1.2 Package Contents

- TR400 Industrial Router × 1
- DC 12V/2A Power Adapter × 1
- Wi-Fi Antenna × 2 + 4G Antenna × 2

- Ethernet Cable (RJ45) × 1
- Quick Start Guide × 1

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Chapter 2 Quick Start

2.1 Default Configuration

Before accessing the router, please note the following default configuration information:

Item	Details
Management IP Address	192.168.0.1
Username	root
Password	000000
WiFi 2.4GHz SSID	TR400-2.4G (Password: 12345678)
Web Protocol	HTTP (Port 80) / HTTPS (Port 443)

Hardware Installation

1. Connect the power adapter to the router and plug it into a power outlet.
2. Use an Ethernet cable to connect your computer to any available LAN port.
3. Attach the external antennas to the antenna connectors.
4. (Optional) Insert a SIM card into the SIM card slot to enable cellular network connectivity.
5. (Optional) Insert an SD card to expand storage capacity.

2.2 Logging into the Management Interface

Steps to access the Web management interface:

1. Open a web browser (Chrome, Firefox, Edge, or Safari recommended).
2. Enter `http://192.168.0.1` in the address bar and press Enter.
3. Enter the username (root) and default password (000000).
4. Click the "Login" button to enter the management dashboard.

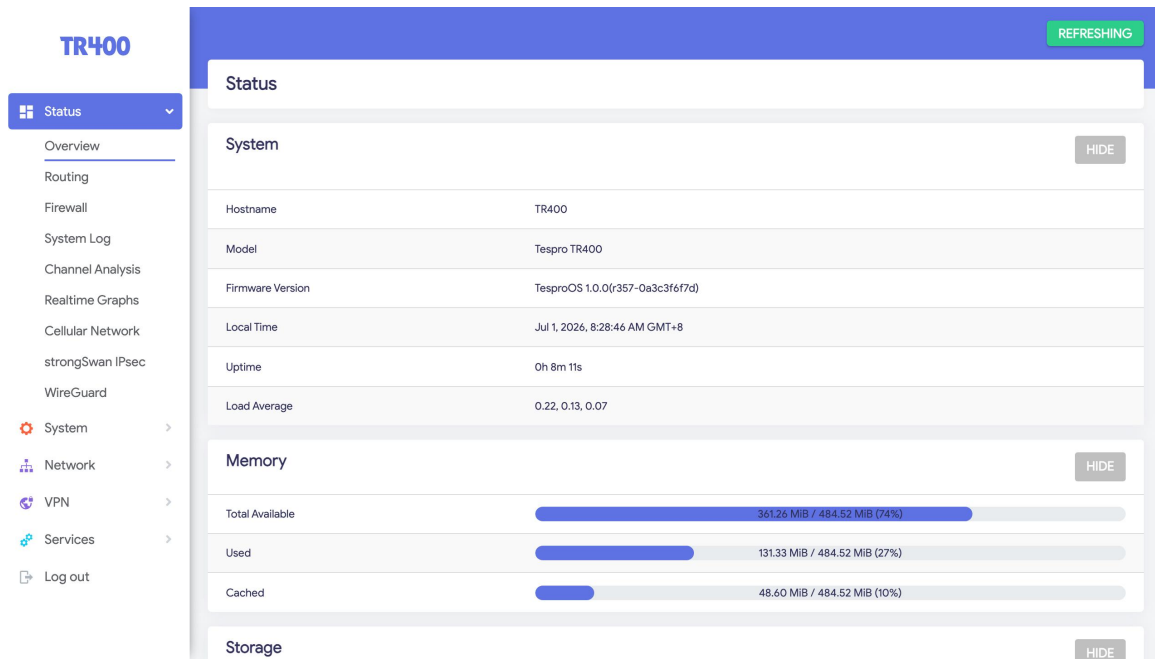


Figure 2-1: Login Page

Warning: Please change the default password immediately after your first login to ensure security. Navigate to System > Administration to set a new password.

2.3 Switching Interface Language

The TR400 supports multiple interface languages. Steps to switch the language:

1. Navigate to System > System > Basic Settings.
2. Click the "Language and Interface" tab.
3. Select your preferred language from the "Language" drop-down menu.
4. Click "Save & Apply" to take effect.

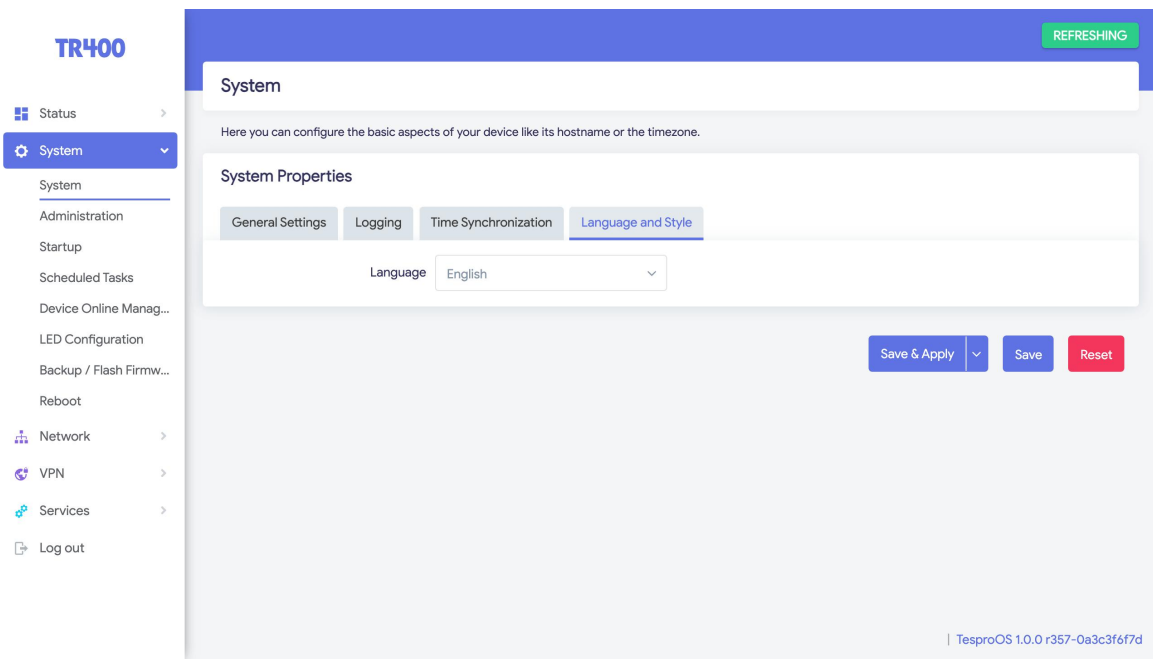


Figure 2-3: Language and Interface Settings

Chapter 3 Status & Monitoring

The Status section provides real-time information on router operational status, network connectivity, and system health. Access these pages through the "Status" menu in the top navigation bar.

3.1 Overview

The Overview page displays a summary of the router's current status, including system information (hostname, model, firmware version, kernel version), memory and storage usage, network interface status, DHCP leases, and connected WiFi clients. This is the default landing page after login.

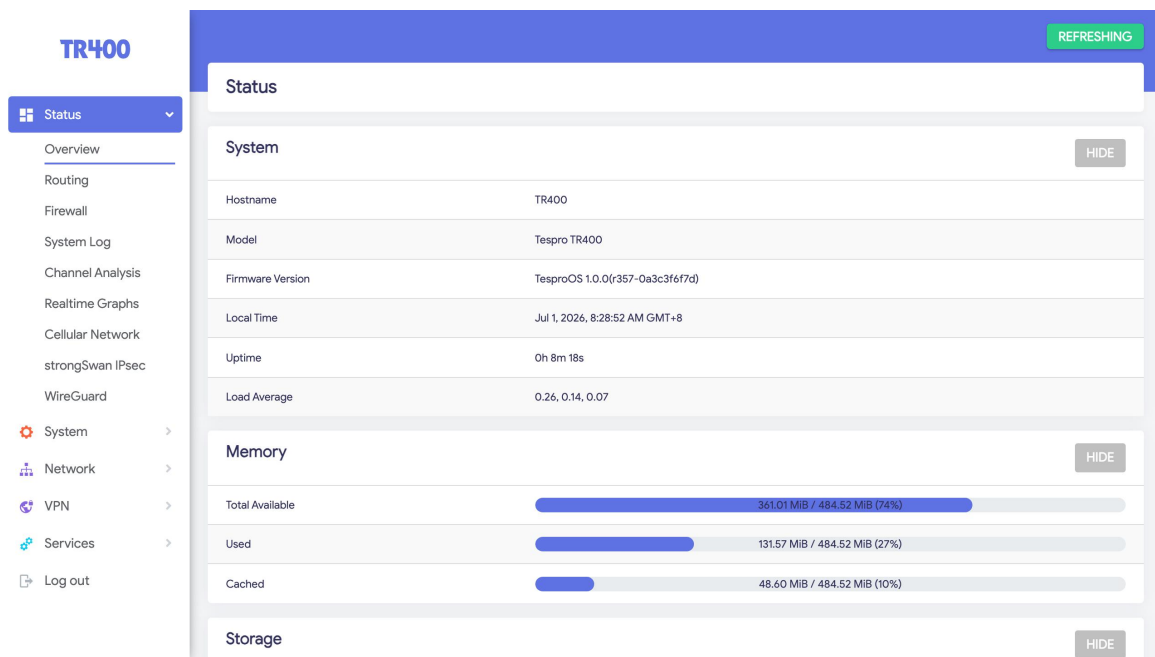


Figure 3-1: Status Overview

3.2 Routing Table

The Routes page displays the current active IPv4 and IPv6 routing tables. Each entry includes the destination network, gateway address, metric, and associated network interface.

TR400

Status

- Overview
- Routing
- Firewall
- System Log
- Channel Analysis
- Realtime Graphs
- Cellular Network
- strongSwan IPsec
- WireGuard
- System
- Network
- VPN
- Services
- Log out

Routing

The following rules are currently active on this system.

IPv4 Routing IPv6 Routing

IPv4 Neighbours

IP address	MAC address	Interface
192.168.0.167	00:EO:4C:2D:4C:10	lan

Active IPv4 Routes

Device	Target	Gateway	Source	Metric	Table	Protocol
wwan	0.0.0.0/0	192.168.0.1	-	0	main	
lan	192.168.0.0/24	-	192.168.0.1	0	main	
wwan	192.168.0.1	-	10.6.81.93	0	main	

Active IPv4 Rules

Priority	Rule
----------	------

Figure 3-2: Routing Table

3.3 Firewall Status

View active IPv4 and IPv6 firewall rules and chains. This page displays all configured rules, packet counters, and byte counters.

TR400

Status

- Overview
- Routing
- Firewall
- System Log
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- strongSwan IPsec
- WireGuard
- System
- Network
- VPN
- Services
- Log out

IPv4/IPv6 traffic table "fw4"

Traffic filter chain "input"

Hook: input (Capture incoming packets routed to the local system), Priority: 0
Policy: drop (Drop unmatched packets)

Rule matches	Rule actions
# Ingress device id is lo	Accept packet
# Verdict map: Conntrack state is established or related	established: Accept packet related: Accept packet
# TCP flags & fin / syn is syn	Continue in syn_flood
# Ingress device name is br-lan	Continue in input_lan
# Ingress device name in set { eth1, ppp-wwan }	Continue in input_wwan
Any packet	Continue in handle_reject

Traffic filter chain "forward"

Hook: forward (Capture incoming packets addressed to other hosts), Priority: 0
Policy: drop (Drop unmatched packets)

Rule matches	Rule actions
# Verdict map: Conntrack state is established or related	established: Accept packet related: Accept packet
# Ingress device name is br-lan	Continue in forward_lan

Figure 3-3: Firewall Status

3.4 System Log

The System Log displays operational messages from system services, network events, authentication attempts, and more. Messages are displayed in chronological order with timestamps.

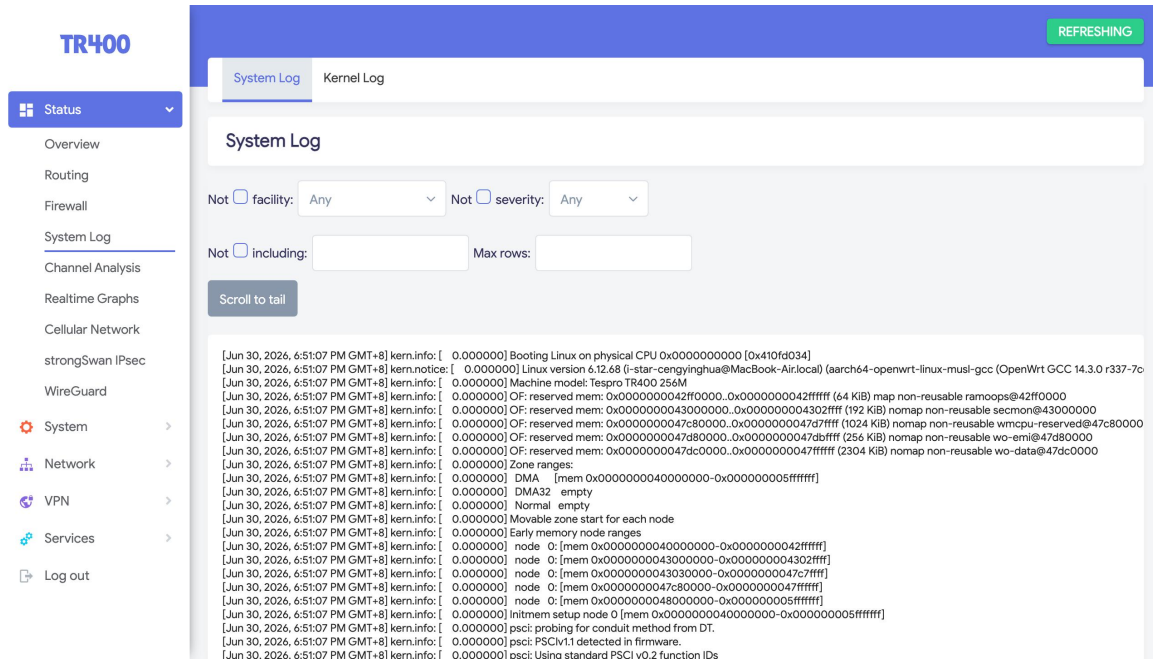


Figure 3-4: System Log

3.5 Kernel Log

The Kernel Log displays messages from the Linux kernel, useful for diagnosing hardware initialization, driver loading, and low-level system events.

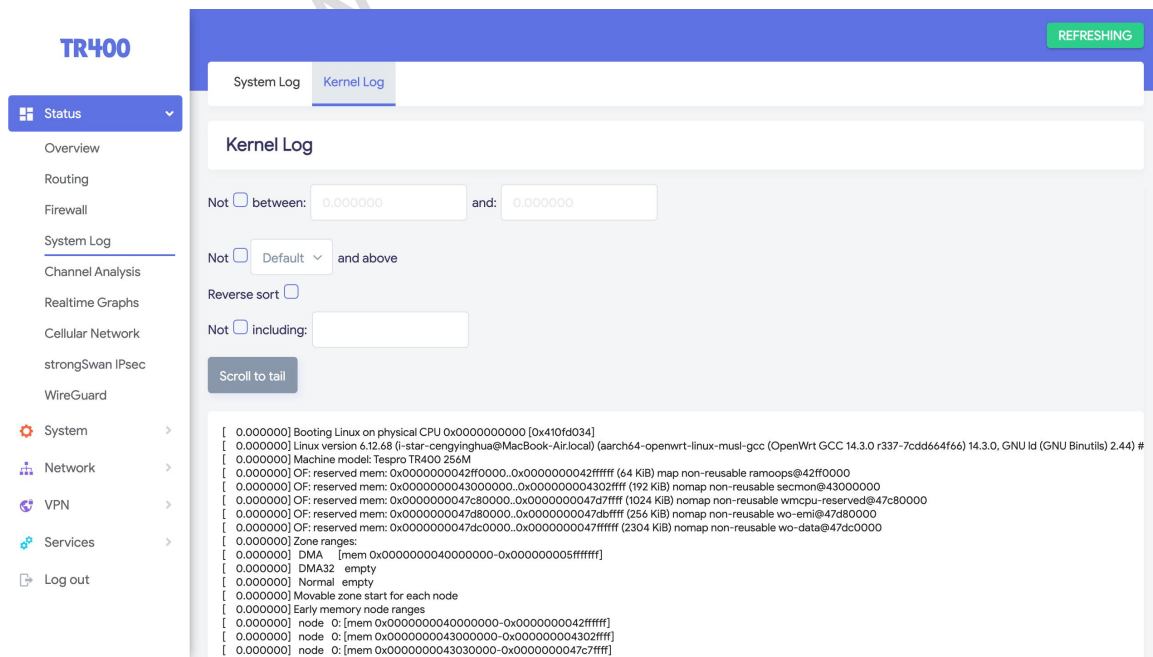


Figure 3-5: Kernel Log

3.6 Channel Analysis

The Channel Analysis tool scans WiFi frequencies to identify channel usage in your environment, helping you select the optimal channel for your WiFi network.

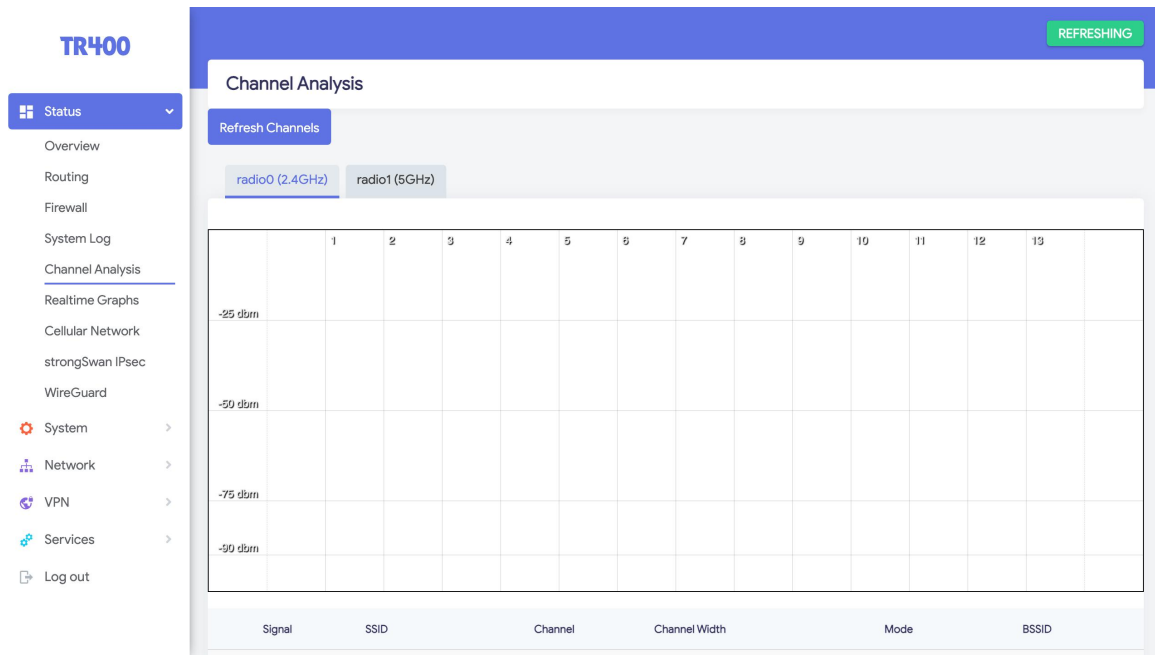


Figure 3-6: WiFi Channel Analysis

3.7 Real-time Graphs

View real-time graphs of network traffic throughput, system CPU load, and active connection statistics.

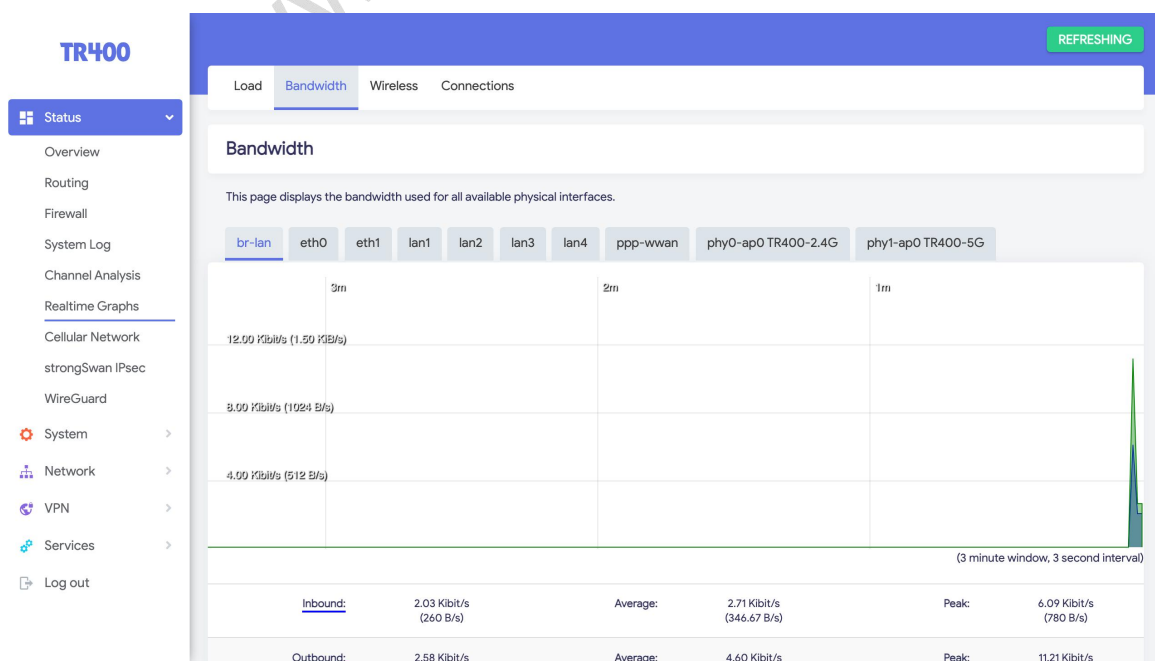


Figure 3-7: Real-time Graphs

3.8 Cellular Network Status

When a 4G/5G cellular module is installed, this page displays the cellular network connection status, including signal strength, carrier information, and connection details. Navigate to Network > Cellular Network for detailed information.

The screenshot shows the TR400 web interface. On the left is a sidebar menu with options: Status (selected), Overview, Routing, Firewall, System Log, Channel Analysis, Realtime Graphs, Cellular Network (highlighted), strongSwan IPsec, WireGuard, System, Network, VPN, Services, and Log out. The main content area is titled 'Cellular Network' and includes a 'REFRESHING' button. Below the title is a tab for 'EC200U'. The page is divided into two sections: 'Modem Info' and 'Network Registration'.

Modem Info	
Manufacturer	Quectel
Model	EC200U
Revision	EC200UEUAAAR03A20M08
IMEI	867963079974813
Device Identifier	7633b1e0df5783808511ca07c891b09054d74772
Power State	on
State	connected

Network Registration	
Mobile Number	
Access Technologies	Lte
Operator	CHN-CT

Figure 3-8: Cellular Network Status

Chapter 4 System Configuration

The System section allows you to configure core router settings, including basic system properties, administrative access, startup services, scheduled tasks, device online management, LED behavior, software updates, and firmware management.

4.1 Basic Settings

Configure basic system settings including device hostname, timezone selection, and NTP server (for automatic time synchronization).

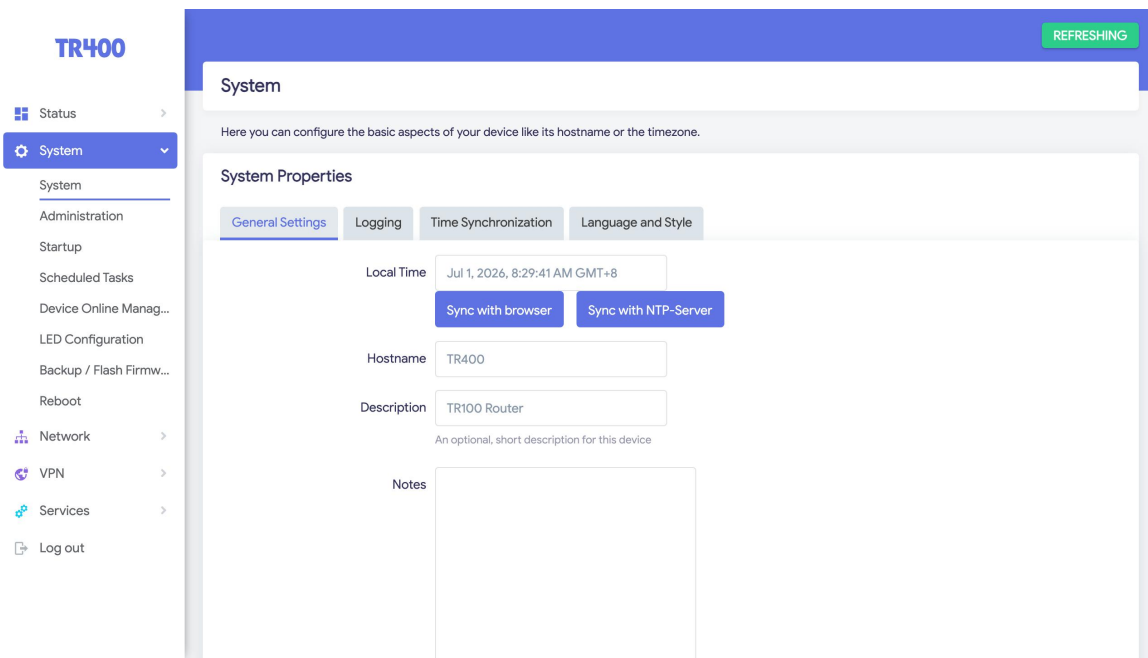


Figure 4-1: Basic System Settings

4.2 Administration

Configure router administration settings including password management, SSH access settings, and Web interface options.

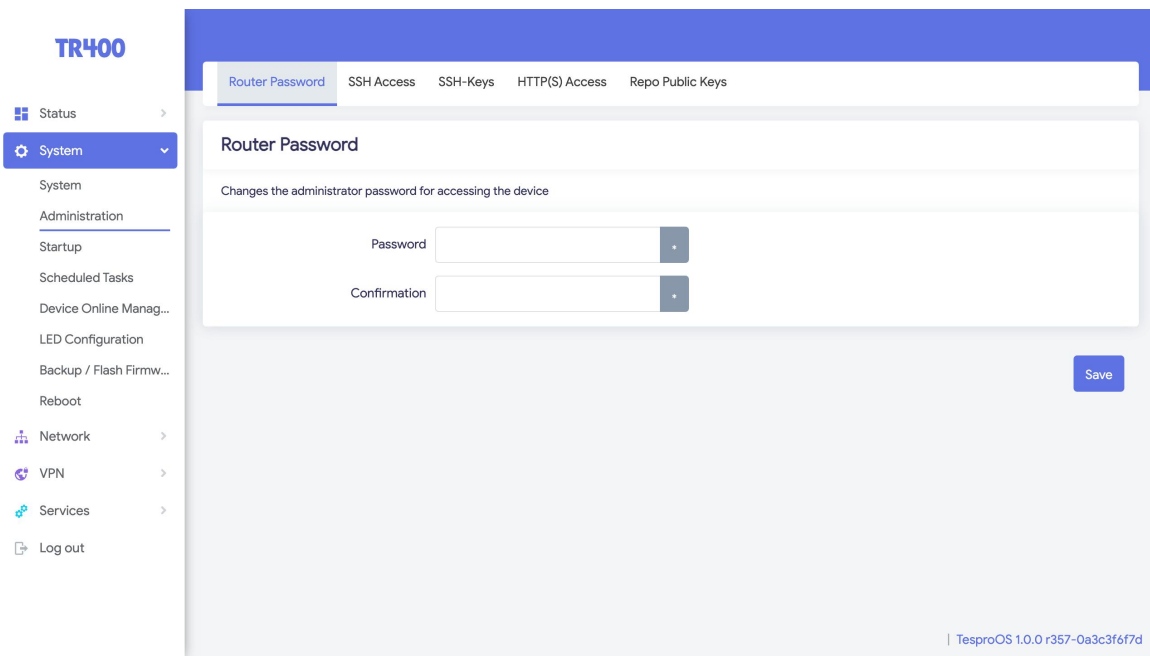


Figure 4-2: Administration Settings

Warning: Always use a strong password. The default password (000000) should be changed immediately after your first login.

4.3 Startup Services

Manage system services and their startup behavior. Enable or disable services and view their current status.

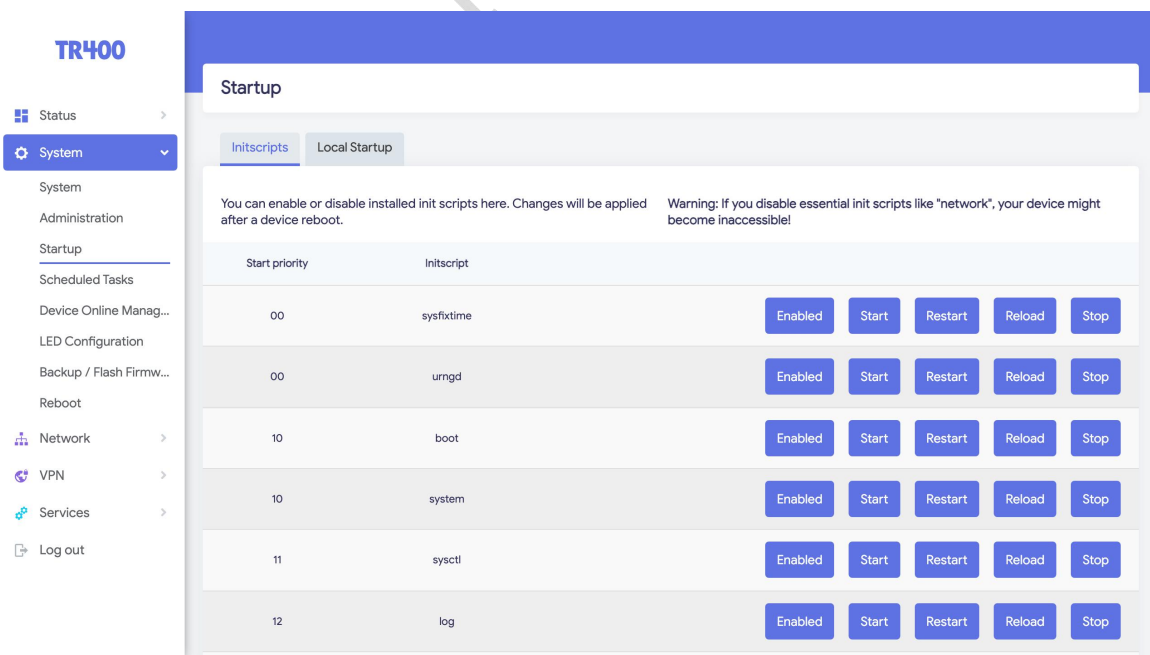


Figure 4-3: Startup Services

4.4 Scheduled Tasks

Configure scheduled tasks using standard cron syntax for automated time-based task execution.

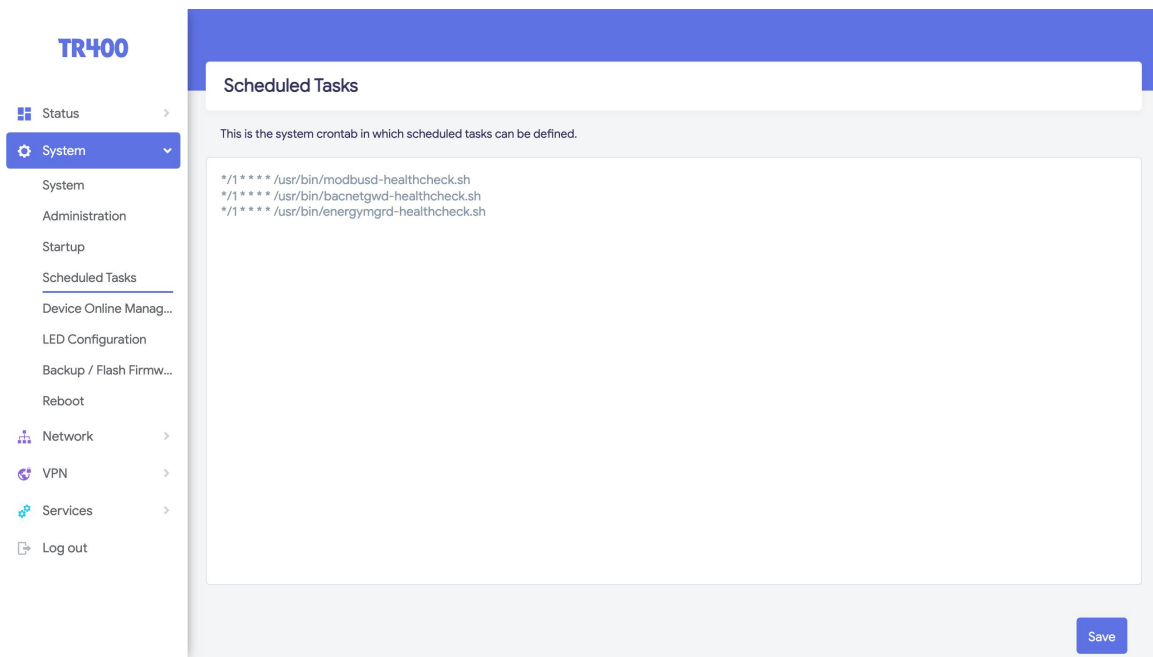


Figure 4-4: Scheduled Tasks (Cron)

4.5 Device Online Manager

The Device Online Manager is a built-in unified device management service on the TR400, providing device status reporting, remote access tunnels, package management, and online firmware updates. By connecting to the Tespro Cloud Management Platform (mgr.tespro.com), you can remotely monitor and manage devices anytime, anywhere.

4.5.1 Settings

The Settings page includes a service status display and four configuration sections. The service status shows the current service running status, cloud connection status, and device information.

- **General Settings:** Enable/disable the Device Online Manager service, set the device name and server address. The device name is used to identify the device on the cloud platform; if left blank, the MAC address will be used.
- **Cloud Management:** Configure MQTT status reporting. When enabled, the device will periodically report its operational status to the cloud server via the MQTT protocol. Configurable parameters include: MQTT broker address and port (default: mgr.tespro.com:8883), TLS encryption (recommended), CA certificate path, authentication username and password, reporting interval (seconds), and reporting only changed data after the initial full data report.
- **Remote Access:** Configure WebSocket tunnels for remote device access. When enabled, the device's Web management interface can be accessed remotely through the cloud platform's Web access URL. Configurable parameters include: tunnel server address and port,

TLS encryption, authentication token, heartbeat interval (seconds), and custom subdomain (defaults to device name).

- **Proxy Services:** Configure the list of local services accessible through the remote tunnel. By default, this includes the Web management interface (HTTP, port 80) and SSH access (TCP, port 22). Service entries can be added, edited, or removed.

Service Status	
Service Status	Running
Cloud Connection	Connected
Device MAC	d60dc4b8775d
Device ID	D60DC4B8775D
Cloud Server	mgrtespro.com:8883
TLS Encryption	Enabled
Remote Access Tunnel	Enabled
Subdomain	d60dc4b8775d (From Device Name)
Web Access URL	http://d60dc4b8775d.mgrtespro.com

Figure 4-5: Device Online Manager - Service Status

4.5.2 Package Management

The Package Management page allows you to install, update, or remove software packages on the device from a cloud repository. The page displays device disk space usage and a list of available packages.

Disk Space	
Total	74.6 MB
Available	69.9 MB
Usage	6%

Software Packages						
Package	Local Version	Server Version	Category	Size	Description	Actions
NetWork Smart Filter oaf	Not installed	6.1.6-r1	系统工具	2.4 MB	NetWork Smart Filter oaf	Install
Kmod-Oaf Of NetWork Smart Filter	Not installed	6.12.68.6.1.6-r1	系统工具	148.1 KB	Kmod-Oaf Of NetWork Sma...	Install

Figure 4-7: Package Management

4.5.3 Firmware Update

The Firmware Update page displays the current device firmware information and a list of firmware versions available from the cloud. You can check online and download the latest firmware for upgrade.

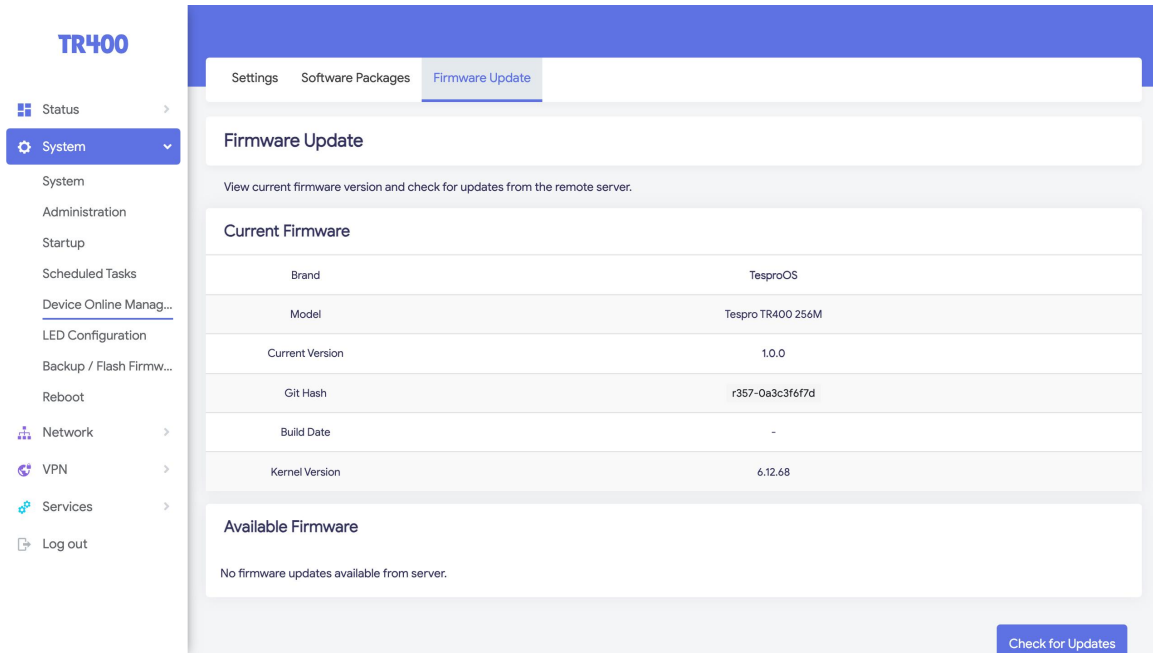


Figure 4-8: Online Firmware Update

Warning: Do not disconnect power or close the browser during a firmware update. The device will automatically restart after the update is complete.

4.6 LED Configuration

Customize device LED indicator behavior by assigning different triggers to each LED.

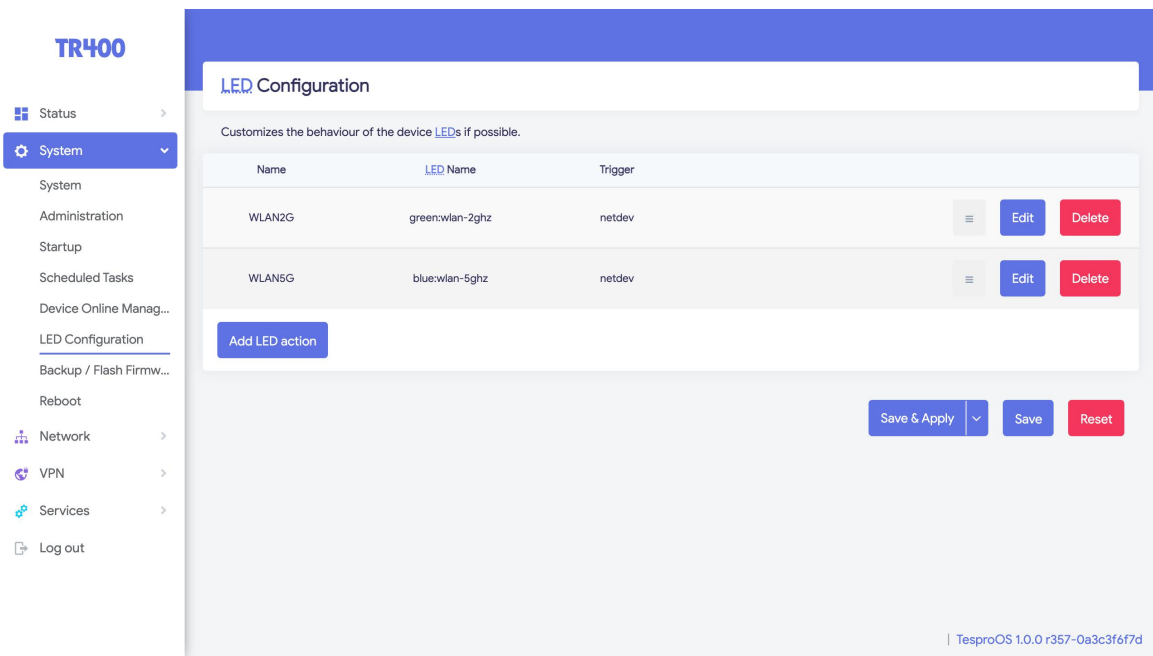


Figure 4-9: LED Configuration

4.7 Backup & Update

4.7.1 Backup / Flash Firmware

Create system backups, restore from backup files, or flash new firmware images.

Tip: Always create a backup before flashing new firmware.

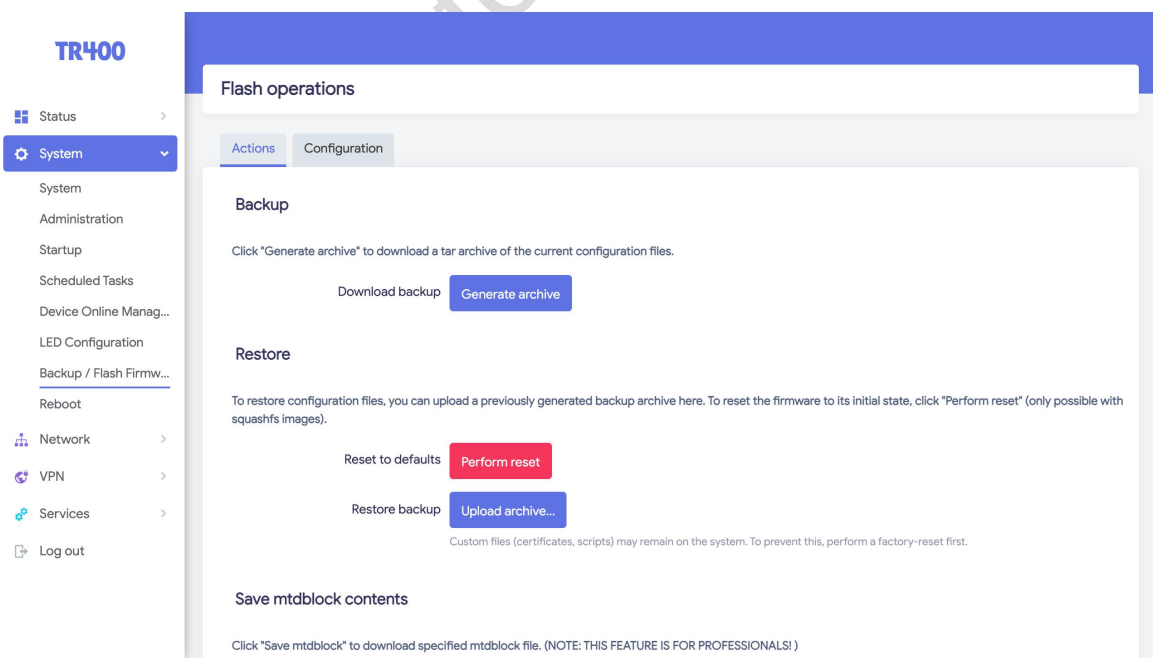


Figure 4-11: Backup & Flash Firmware

Warning: Do not disconnect power during a firmware upgrade. This may permanently damage the device.

4.7.2 Configuration Management

Import or export device configuration files for backup or multi-device deployment.

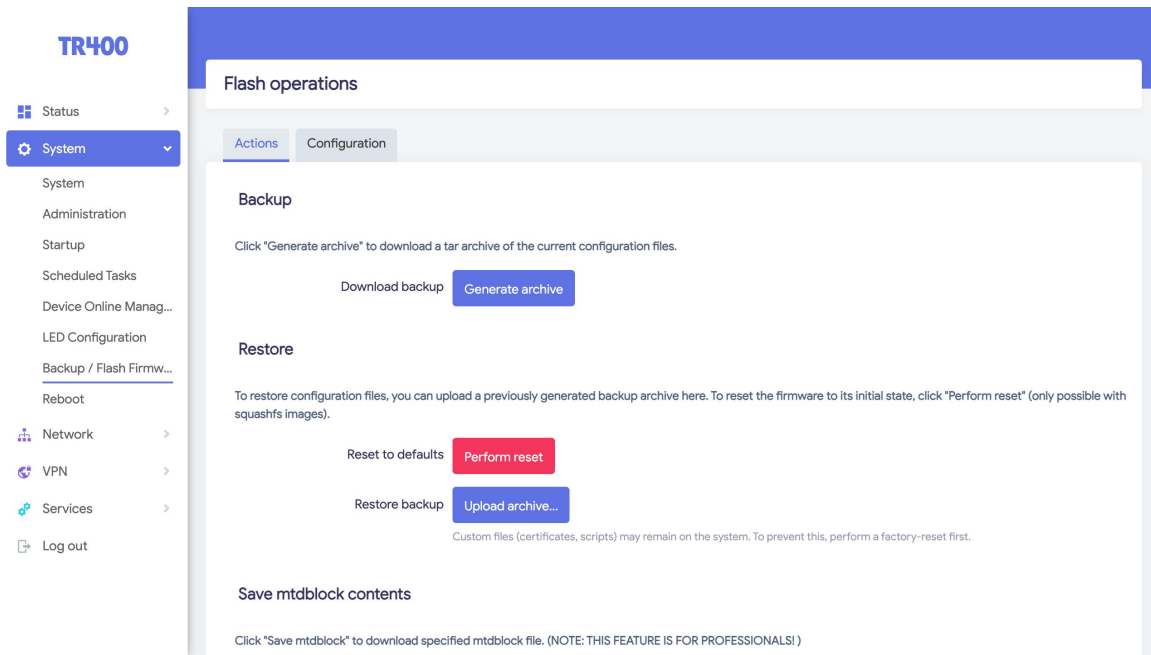


Figure 4-12: Configuration Management - Backup & Restore

4.8 Reboot

Perform a controlled system reboot. The router will gracefully shut down all services and restart.

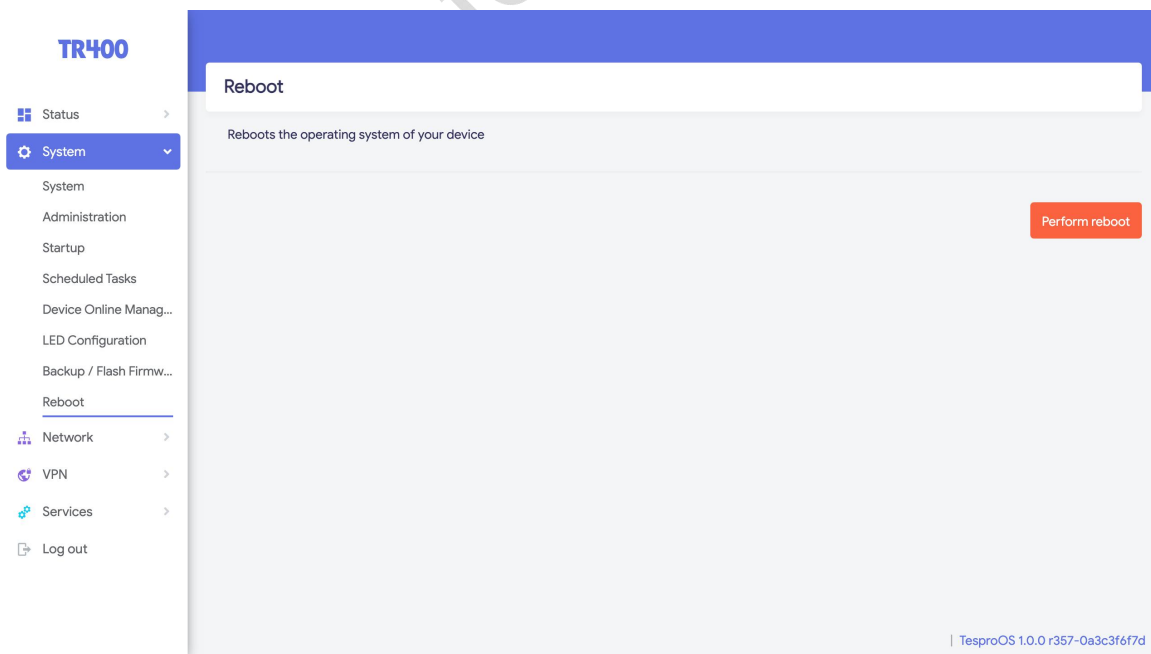


Figure 4-14: System Reboot

Chapter 5 Network Configuration

The Network section provides comprehensive control over all network-related settings.

5.1 Network Interfaces

Configure network interfaces including LAN, WAN, and cellular network (WWAN) connections. Each interface can be configured with independent protocols, IP addresses, and parameters.

Tip: The LAN interface is configured in bridge mode (br-lan) by default, combining all LAN Ethernet ports into a single network segment. The TR-22X provides 2 LAN ports, and the TR-24X provides 4 LAN ports.

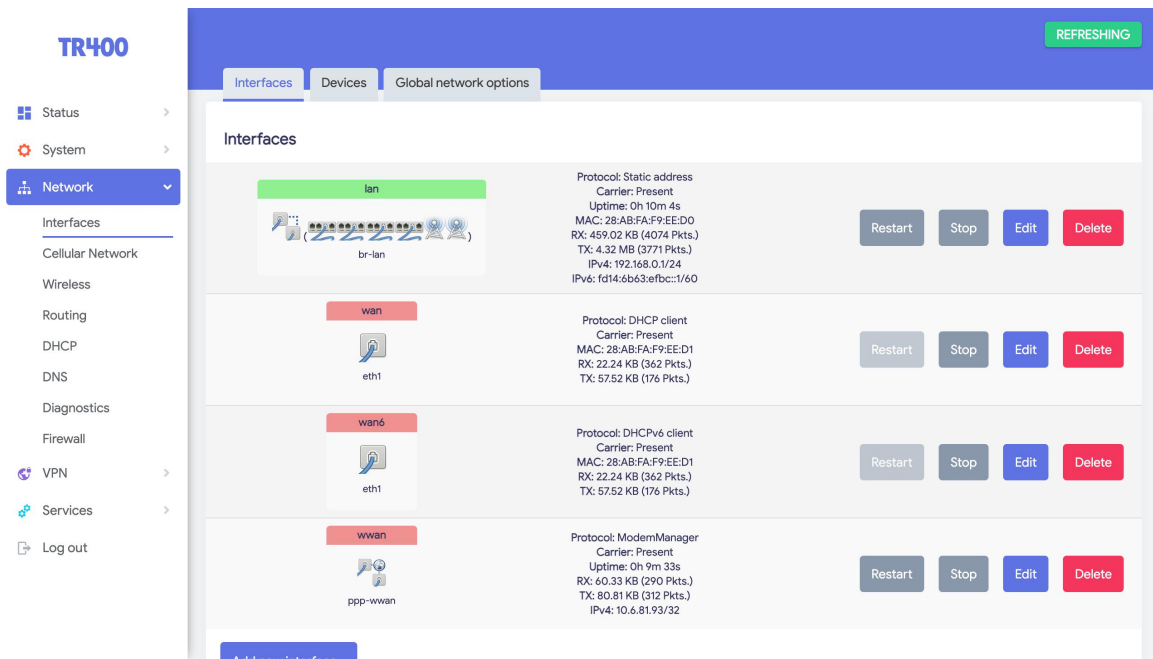


Figure 5-1: Network Interfaces

5.2 Cellular Network Management

Manage cellular module SIM card settings. View SIM status, signal quality, and configure cellular parameters. Supports SIM card slot management, switching policies, and eSIM configuration.

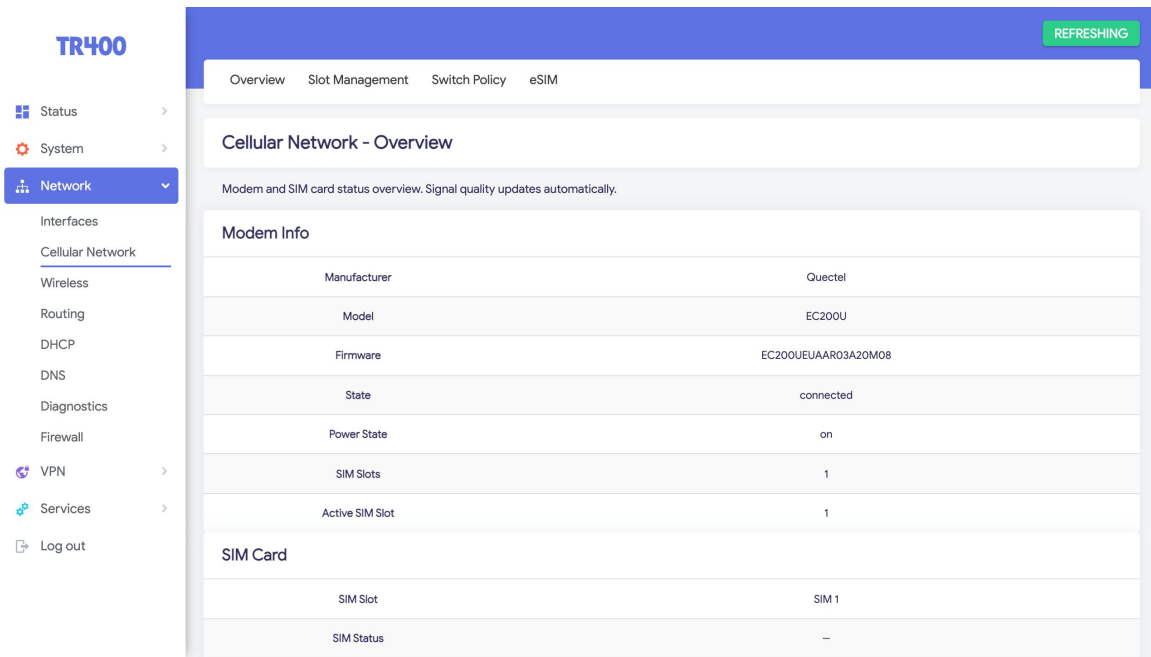


Figure 5-2: Cellular Network Management

5.3 WiFi Network

Configure WiFi settings for the 2.4GHz band, including SSID, encryption method, channel, and transmit power.

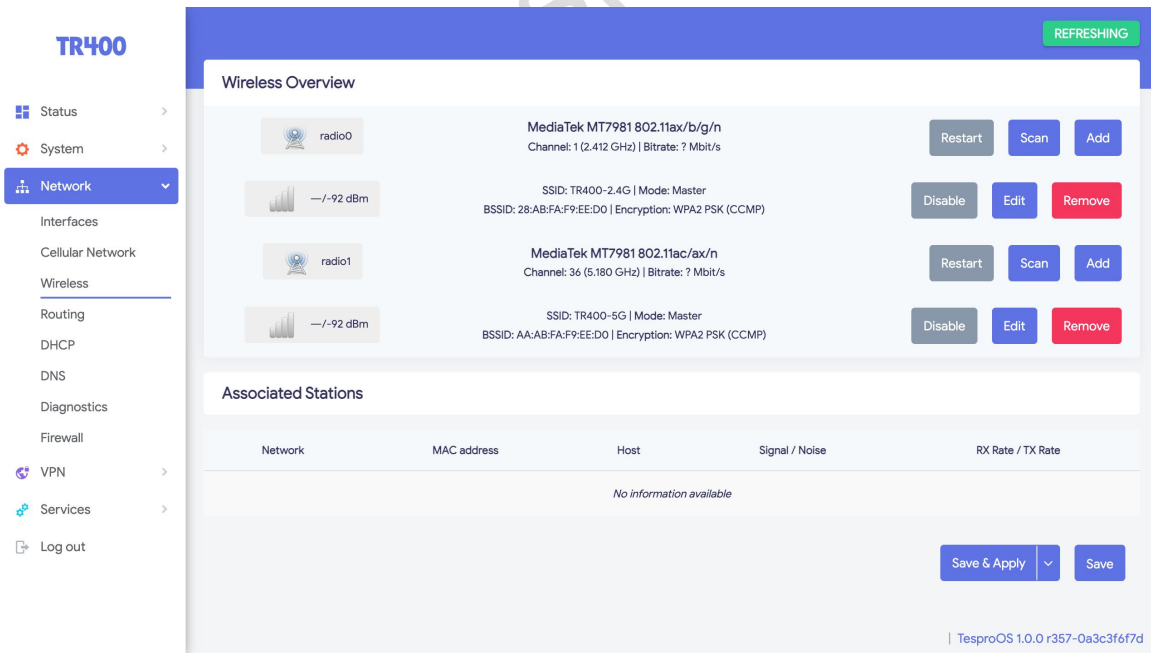


Figure 5-3: WiFi Network Configuration

Warning: Change the default WiFi password after initial setup to prevent unauthorized access.

5.4 Static Routes

Add or manage static route entries to define custom paths to specific destinations for network traffic.

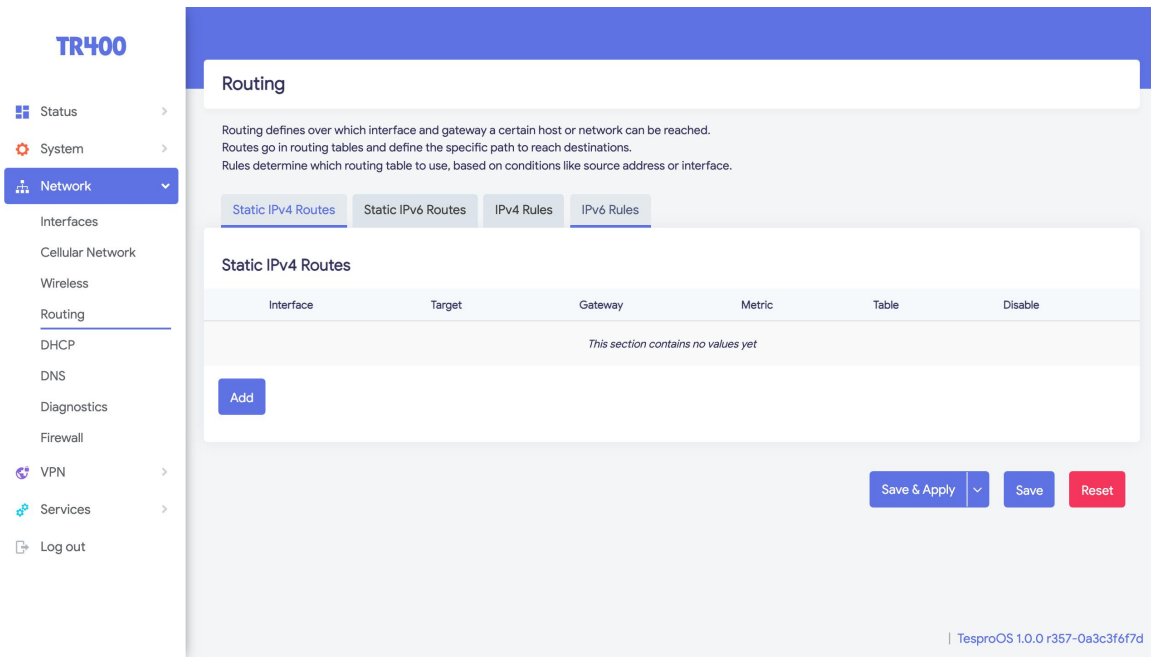


Figure 5-4: Static Routes

5.5 DHCP

Configure the DHCP server for automatic IP address assignment, manage static leases, and view active leases.

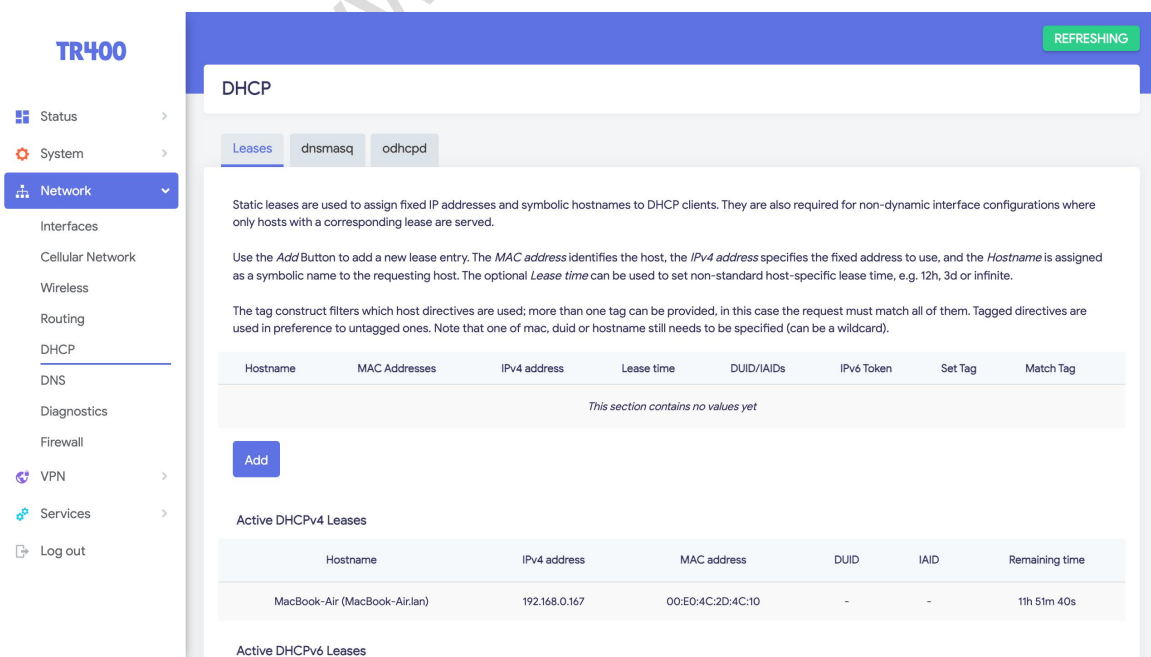


Figure 5-5: DHCP Configuration

5.6 DNS

Configure DNS settings including local domain, DNS forwarding, caching, and custom DNS records.

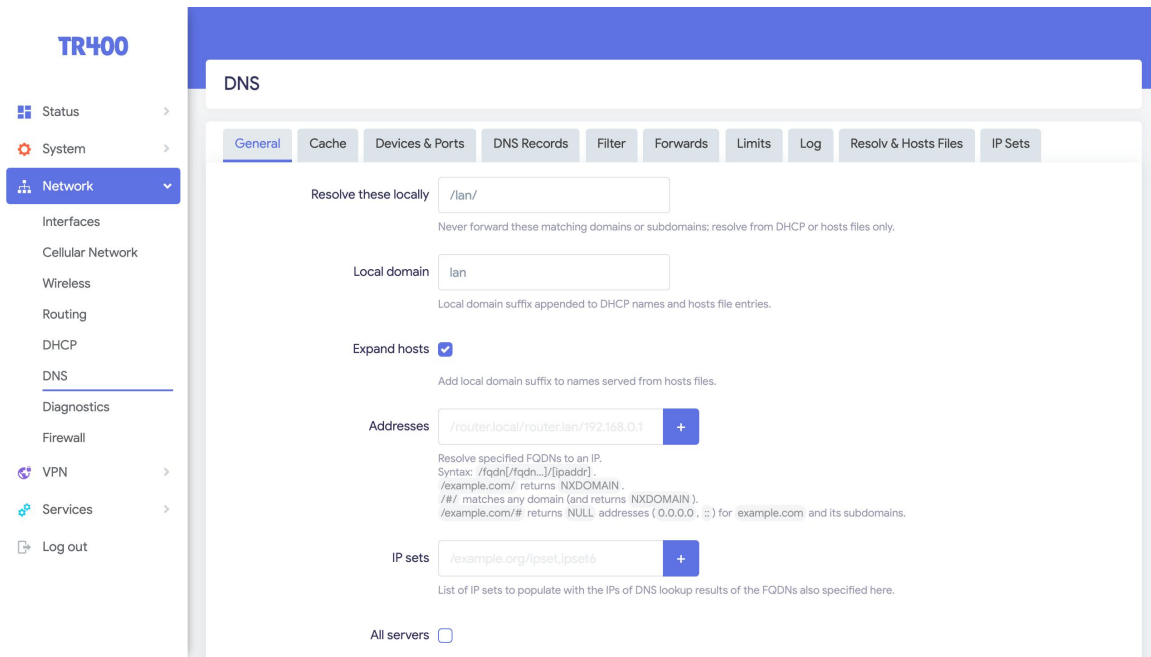


Figure 5-6: DNS Settings

5.7 Network Diagnostics

Built-in network diagnostic tools including Ping, Traceroute, and Nslookup for troubleshooting connectivity issues.

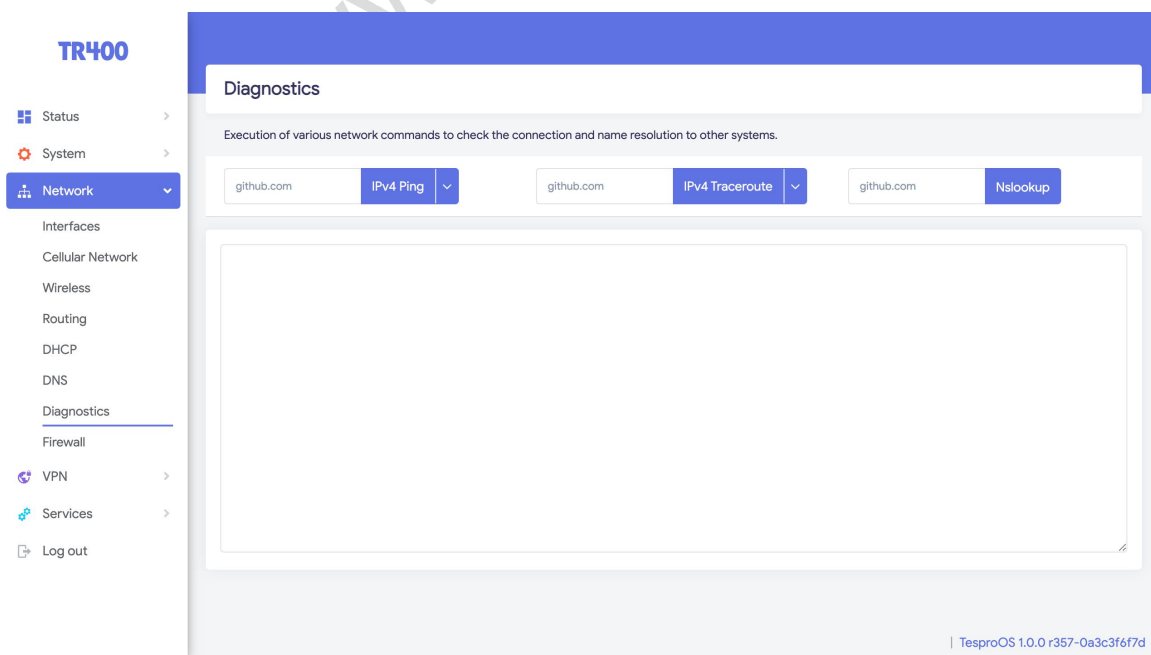


Figure 5-7: Network Diagnostics

Chapter 6 Firewall

The TR400 features a built-in industrial-grade firewall based on nftables (firewall4) with a zone-based architecture.

6.1 General Settings (Zone Configuration)

Configure firewall zones and their default security policies. The TR400 comes pre-configured with LAN (allow) and WAN (deny input, IP masquerading enabled) zones.

Tip: SYN flood protection is enabled by default to defend against denial-of-service attacks.

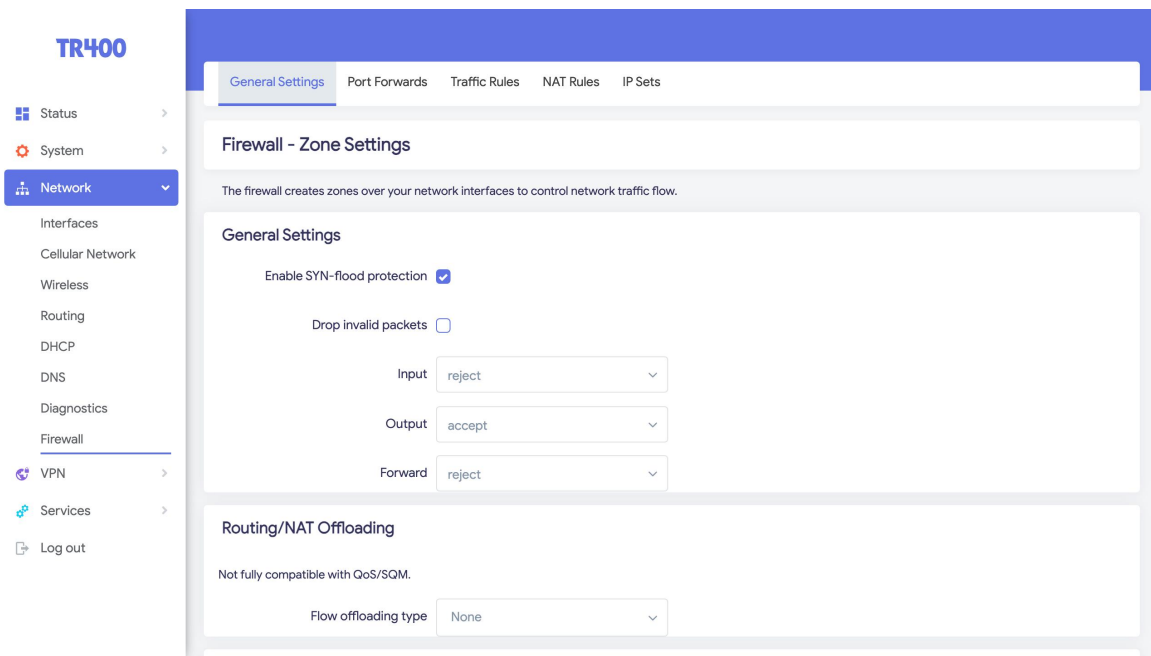


Figure 6-1: Firewall Zone Settings

6.2 Port Forwarding

Configure port forwarding rules to allow external access to services on the LAN.

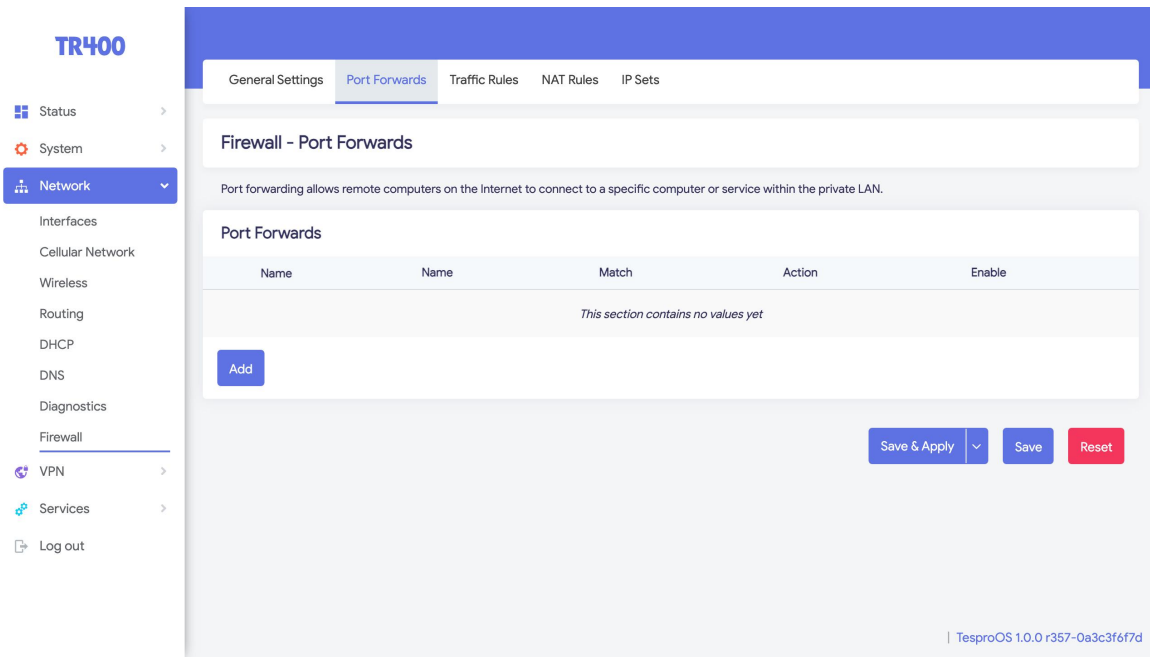


Figure 6-2: Port Forwarding

6.3 Traffic Rules

Define custom firewall rules to allow or block specific traffic. Pre-configured rules include DHCP renewal, Ping (ICMP), IPv6 support, and IPSec VPN passthrough.

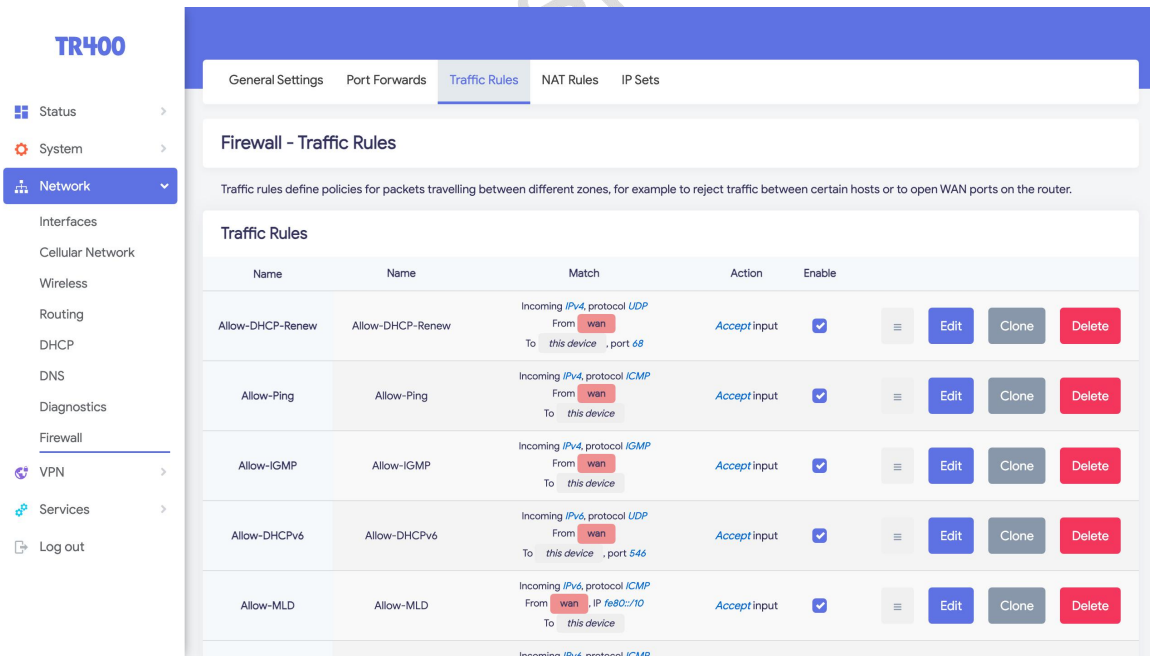


Figure 6-3: Traffic Rules

6.4 NAT Rules

Configure Source NAT (SNAT) rules for fine-grained control over the source IP address used by outbound traffic.

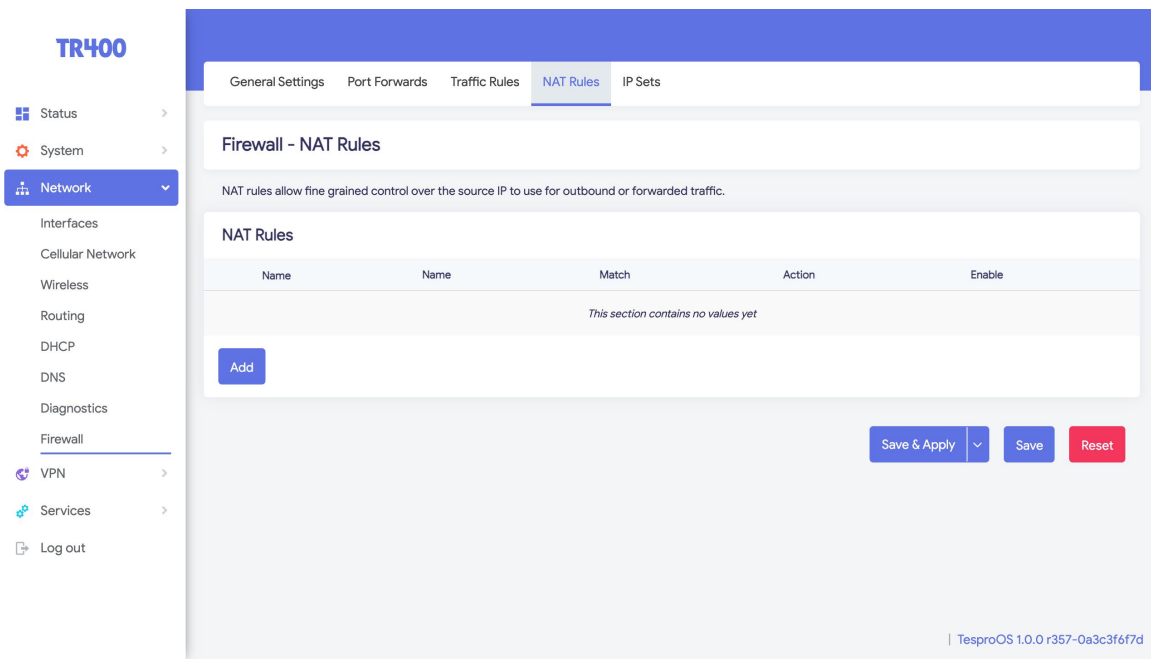


Figure 6-4: NAT Rules

6.5 IP Sets

Create and manage IP sets for efficient matching of large address groups in firewall rules.

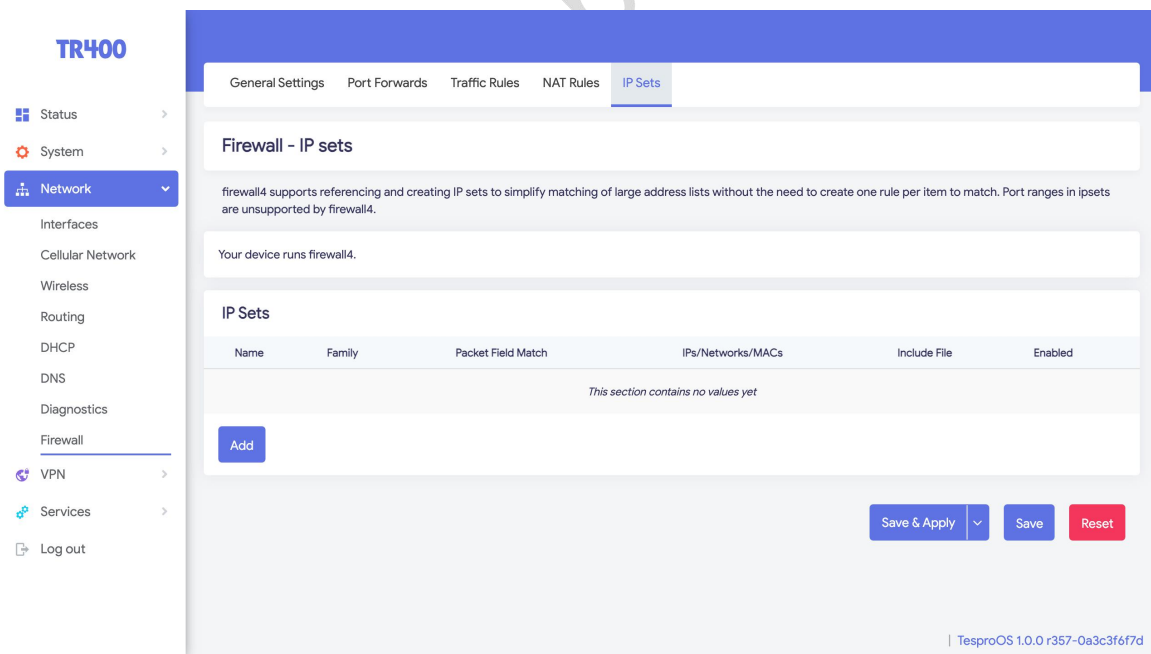


Figure 6-5: IP Sets

Chapter 7 Services

The Services section provides professional service features of the TR400, including serial port communication services and Modbus protocol gateway functions, suitable for industrial and IoT applications.

7.1 Serial Service - Basic Settings

The TR400 features a built-in serial communication service with UDP multicast support. The router provides 3 configurable serial ports:

- RS485-1: /dev/ttyS1
- RS485-2: /dev/ttyS2
- RS232: /dev/ttyS0

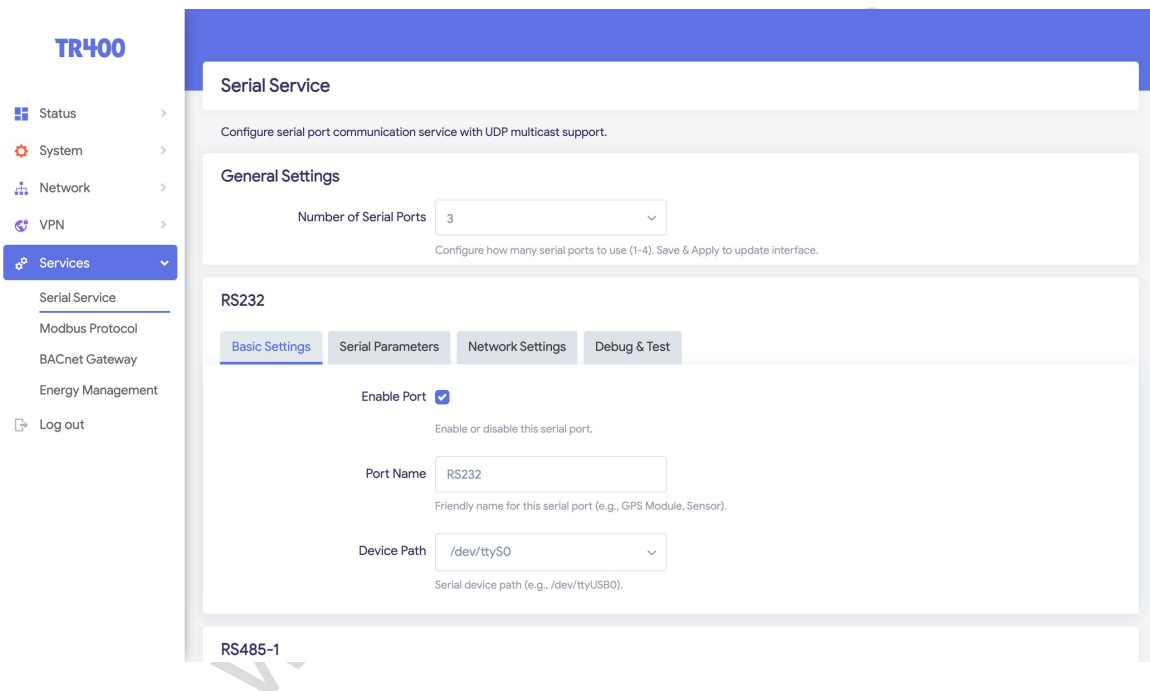


Figure 7-1: Serial Service - Basic Settings

7.2 Serial Parameters

Configure communication parameters for each serial port:

Tip: The router serial port and connected device must use identical communication parameters.

- Baud Rate: Communication speed (300 to 115200 bps), default 9600
- Parity: Error detection method (None, Even, Odd), default None
- Data Bits: Number of data bits per character (6, 7, or 8), default 8
- Stop Bits: Number of stop bits (1, 1.5, or 2), default 1

7.3 Serial Network Settings

Configure UDP multicast network settings for each serial port:

- Multicast Address: UDP multicast group address (default 239.0.0.x)
- Send/Receive/Control Port: Network ports for data and management
- Lock Timeout: Timeout for automatic lock release (seconds)

7.4 Serial Debug & Test

Built-in serial communication debugging and testing interface. Supports sending/receiving data in ASCII or HEX format, real-time adjustment of serial settings, and communication log viewing.

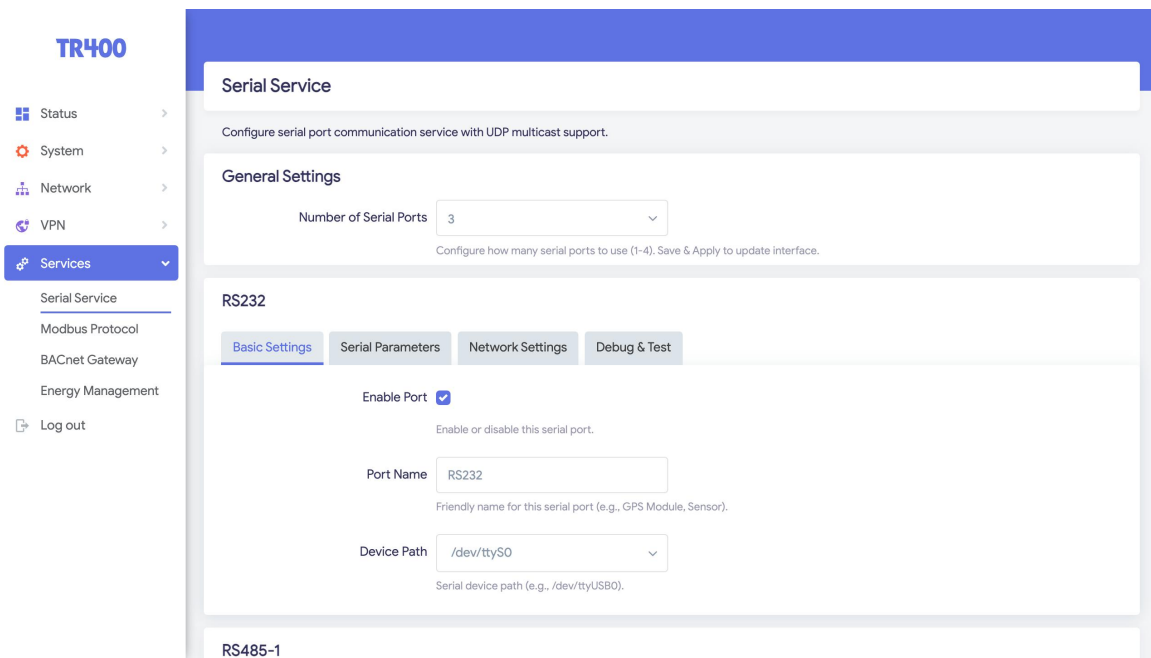


Figure 7-4: Serial Debug & Test

7.5 Modbus Protocol Service - Overview

The TR400 features a fully functional Modbus protocol gateway service that bridges industrial Modbus devices with modern IT/cloud systems. The service supports dual-role mode: as a Master, it polls and collects data from slave devices via southbound serial or TCP connections; as a Slave, it accepts access from external masters and responds to register read/write requests. Northbound channels support six data publishing methods: MQTT, TCP, UDP, HTTP, OPC UA Client, and OPC UA Server.

Tip: The Modbus service is integrated with the serial service module. Serial parameters are configured in the Serial Service section (7.2).

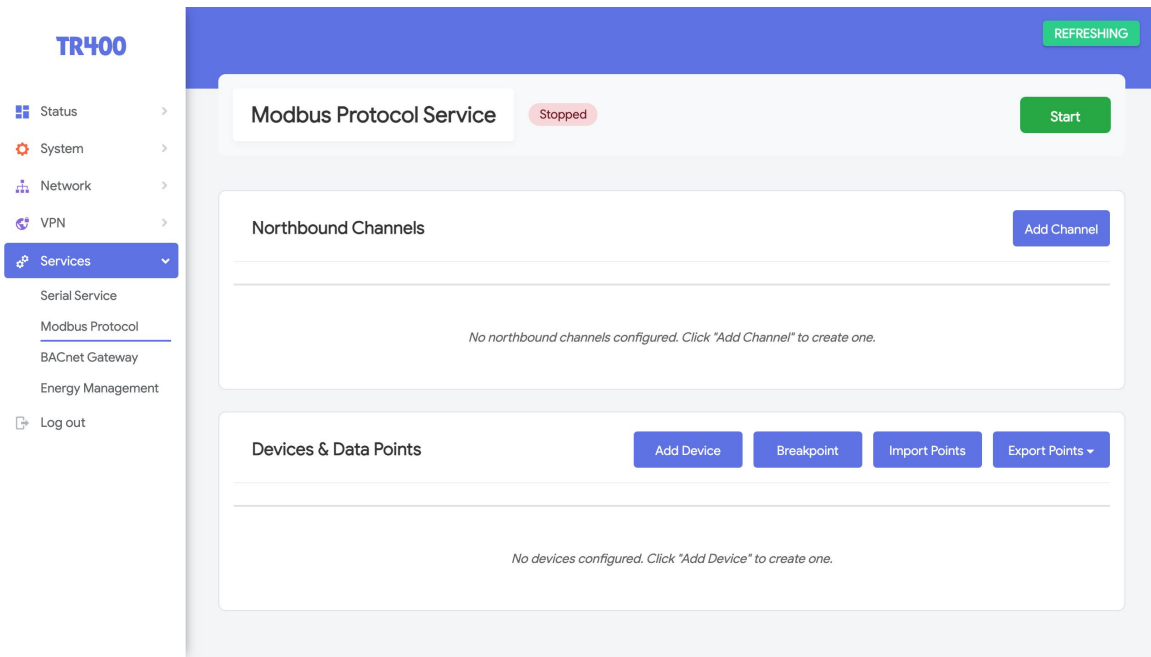


Figure 7-5: Modbus Protocol Service Overview

7.6 Northbound Channels

Northbound channels define how the TR400 forwards collected Modbus data to upstream systems. Six channel types are supported:

- MQTT: Publish data to a message broker via the MQTT protocol.
- TCP: Send data via direct TCP Socket connection.
- UDP: Send data via UDP packets.
- HTTP: Report data via HTTP REST API.
- OPC UA Client: Write data to remote OPC UA server nodes as an OPC UA client.
- OPC UA Server: Act as an OPC UA server, mapping collected data to OPC UA nodes for subscription by external clients.

Multiple channels can be configured simultaneously to send data to different destinations.

7.7 Devices & Data Points

The Devices & Data Points section allows you to define Modbus devices and their register mappings. In Master mode, each device includes slave address, polling interval, data point configuration, and southbound connection settings; in Slave mode, you can configure the TR400's own slave address and externally exposed register mappings, supporting four register types: AO (Holding Register), DO (Coil), AI (Input Register), and DI (Discrete Input).

7.8 Modbus Quick Start Guide

- Step 1: Configure serial parameters in the Serial Service.
- Step 2: Create a northbound channel (e.g., MQTT connection to a cloud platform).
- Step 3: Add Modbus slave devices.
- Step 4: Add data points to map registers.

Step 5: Configure southbound connections.

Step 6: Start the Modbus service.

Warning: Ensure that the serial parameters in the Serial Service exactly match those of the Modbus slave device. Mismatched baud rates or parity settings will cause communication failures.

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Chapter 8 VPN Connection Management

The TR400 supports multiple VPN protocols for secure remote network connectivity. The VPN menu allows you to configure and manage OpenVPN, WireGuard, IPsec (strongSwan), and other VPN services. VPN features are provided as software packages, and you can install the corresponding VPN Bundle as needed.

8.1 VPN Menu Overview

In the left navigation bar of the Web management interface, the VPN menu provides configuration entry points for each VPN protocol. Installed VPN services are displayed in the menu.

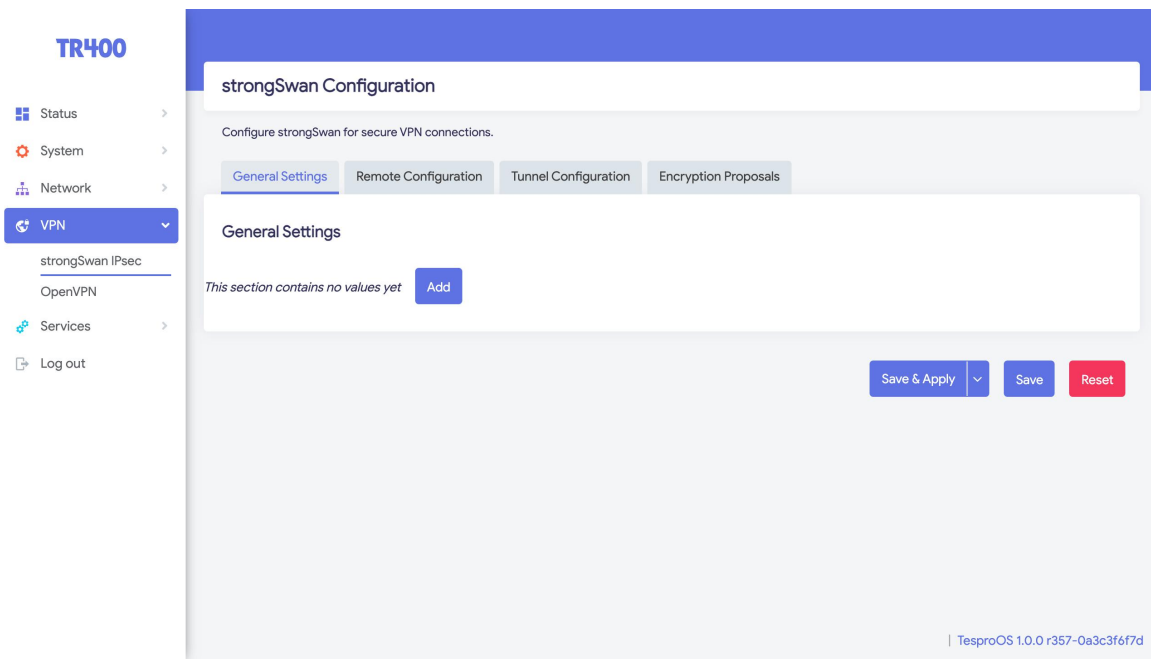


Figure 9-1: VPN Menu Overview

8.2 OpenVPN

OpenVPN is a mature open-source VPN protocol supporting TLS encryption and multiple authentication methods. The TR400 provides the openvpn-bundle package; once installed, you can configure OpenVPN clients or servers under the VPN menu.

Key configuration items include:

- VPN instance name and type (client/server)
- Remote server address and port
- Protocol selection (UDP/TCP)
- TLS certificate and key configuration
- Encryption algorithm selection
- Log level settings

Supports uploading .ovpn configuration files for quick VPN instance creation.

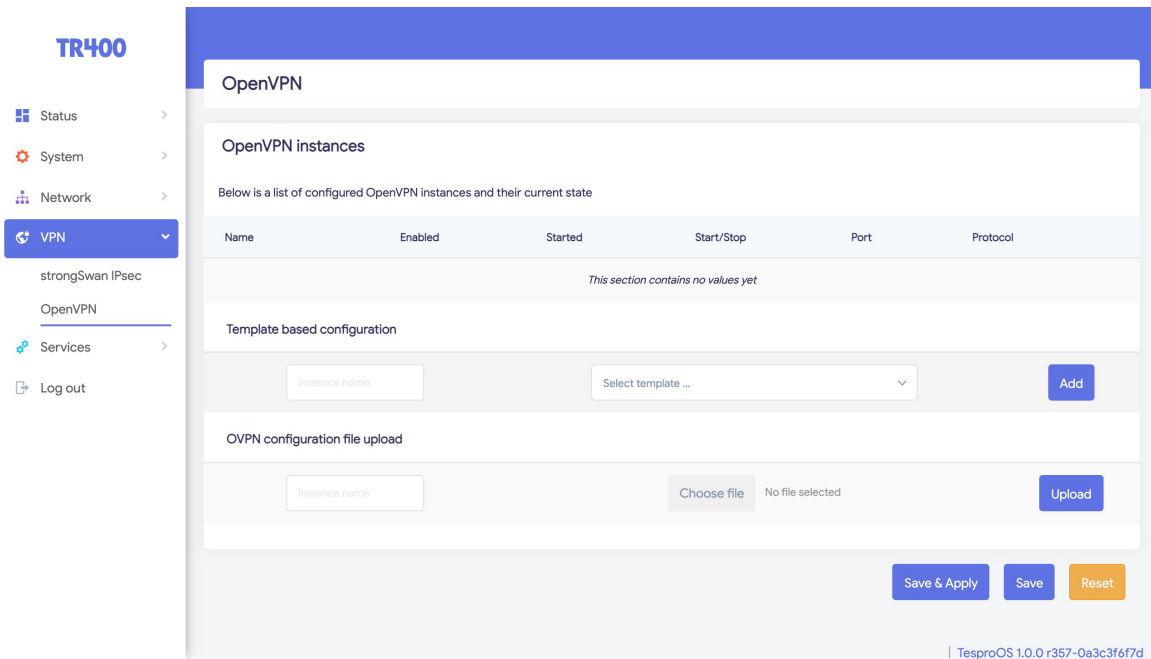


Figure 9-2: OpenVPN Configuration

8.3 WireGuard

WireGuard is a modern high-performance VPN protocol known for its lean codebase and speed. The TR400 provides the wireguard-bundle package. WireGuard configuration is managed through the network interface.

Key configuration items:

- Private and public key generation
- Peer configuration: public key, allowed IP address ranges
- Listening port
- Persistent keepalive interval

WireGuard status can be viewed on the Status > WireGuard page, showing current connection information, transfer statistics, and latest handshake time.

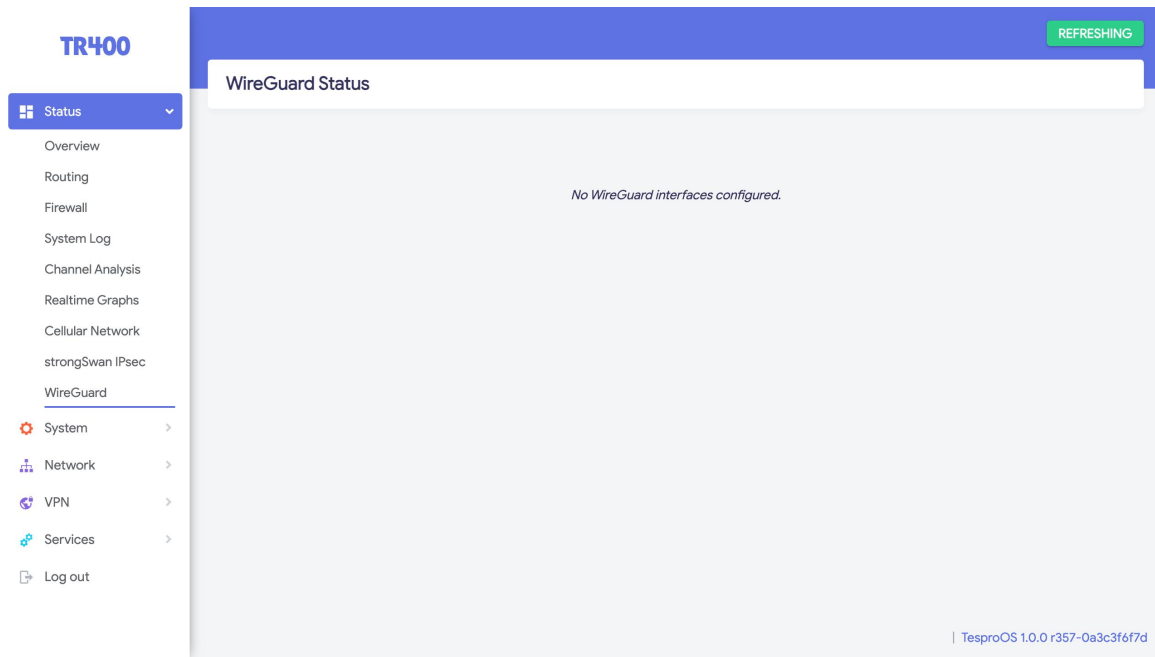


Figure 9-3: WireGuard Configuration

8.4 IPsec (strongSwan)

IPsec is an industry-standard VPN protocol suitable for enterprise-grade site-to-site secure connections. The TR400 provides the ipsec-bundle package, implemented based on strongSwan. Key configuration items:

- IKE version selection (IKEv1/IKEv2)
- Local and remote authentication methods (pre-shared key/certificate)
- Encryption proposal configuration (encryption algorithm, integrity algorithm, DH group)
- ESP proposal configuration
- Local and remote subnets
- IKE and ESP lifetimes

IPsec connection status can be viewed on the Status > strongSwan page.

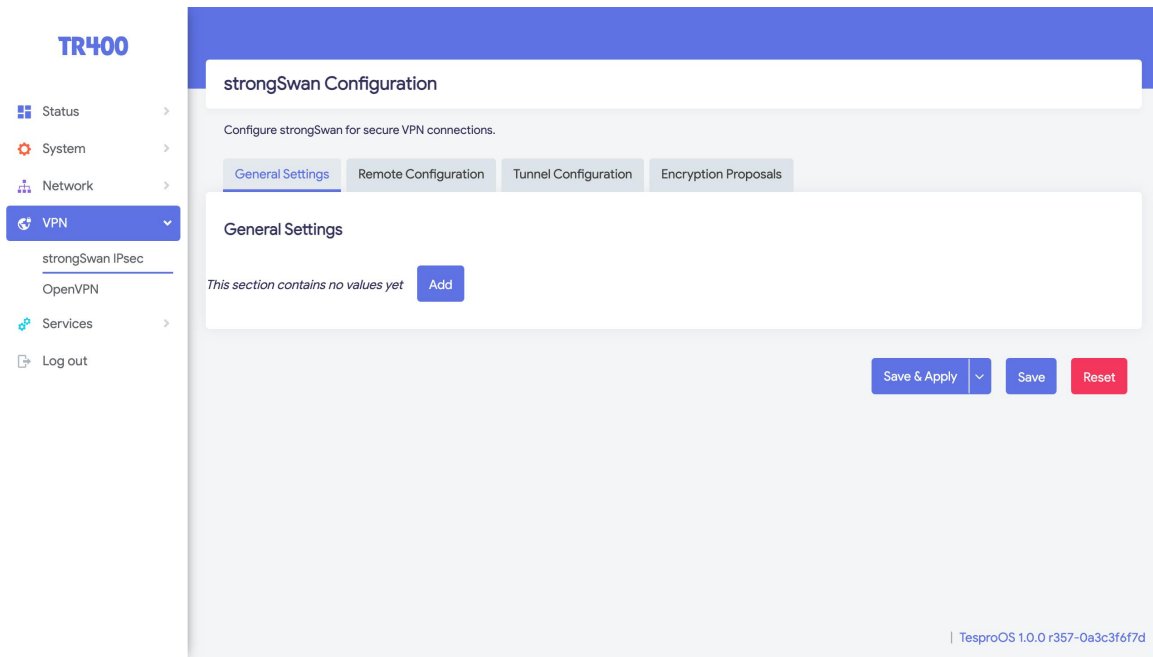


Figure 9-4: IPsec Configuration

8.5 PPTP

PPTP is a legacy VPN protocol with simple configuration but lower security. The TR400 provides the pptp-bundle package. PPTP is managed through UCI configuration files and does not provide a dedicated Web interface.

Configuration method: Access the device via SSH, edit the `/etc/config/pptp` configuration file, or manage PPTP service status through the System > Startup Services interface.

Tip: Due to the lower security of the PPTP protocol, it is recommended to use OpenVPN or WireGuard for VPN connections.

Chapter 9 BACnet Gateway Service

The BACnet Gateway Service collects data from BACnet/IP or BACnet MS/TP devices in building automation systems to the TR400 gateway and forwards it to upper-level systems through multiple channels. It also provides a standard Modbus TCP Server interface for direct read/write access by SCADA systems.

9.1 Service Overview

The BACnet Gateway Service page features a service status bar at the top showing the running status and start/stop buttons; the lower section is divided into two main areas: northbound channel configuration and device & data point management.

Key features:

- Southbound data collection from BACnet/IP and BACnet MS/TP (RS-485) devices
- Northbound reporting to cloud platforms via MQTT/TCP/UDP/HTTP
- Built-in Modbus TCP Server with automatic mapping of BACnet objects to Modbus registers
- BACnet WriteProperty reverse control support
- Breakpoint resume and change-of-value reporting
- CSV/XLSX batch import/export of data point configurations

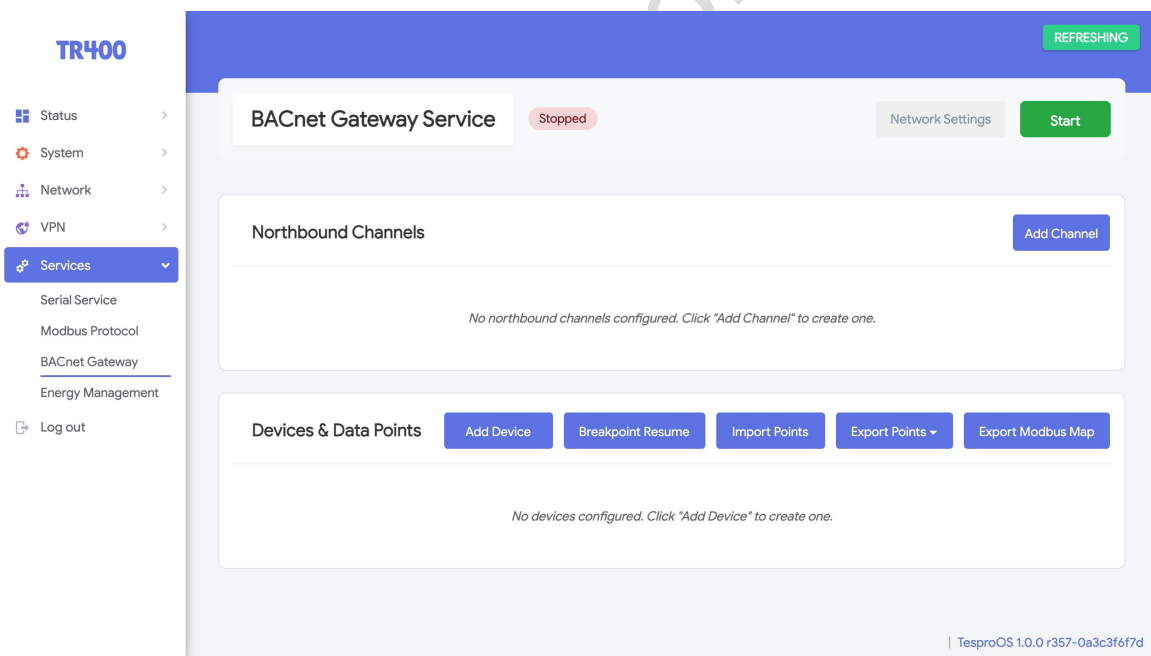


Figure 10-1: BACnet Gateway Service Overview

9.2 Service Start/Stop

Click the Start button to launch the service; the status indicator turns green showing Running. To stop the service, click the Stop button.

9.3 Northbound Channel Configuration

Northbound channels are used to report collected data to remote servers. Multiple channels can be configured simultaneously for parallel operation.

Channel types:

- MQTT: Publish data to an MQTT broker
- HTTP: Send data to an HTTP/HTTPS endpoint
- TCP/UDP: Send data via raw sockets
- TCP Server: TR400 acts as a passive server listening for connections

MQTT channel configuration items: server address and port, topic, client ID, QoS, username/password, TLS enable, reporting interval, data format (JSON/binary).

9.4 Device Management

Click "Add Device" to define a new BACnet slave device:

- Device name (required, must be unique)
- Modbus Slave ID (1-247)
- BACnet instance number
- Southbound connection type (BACnet/IP or BACnet MS/TP)

Supported operations: save device, delete device, control panel, report format, copy device.

9.5 Southbound Connection Configuration

BACnet/IP mode: Configure the target IP address, port (default 47808), timeout, and retry count.

BACnet MS/TP mode: Configure serial port selection, local MAC address, device MAC address, and maximum master number.

Tip: Serial port baud rate and other parameters are configured centrally in the Serial Service module.

9.6 Data Point Management

Data points define BACnet object mappings within a device:

- Name and report key
- BACnet object type (AI/AO/AV/BI/BO/BV/MSI/MSO/MSV)
- Object instance number
- Data type (float32/int16, etc.)
- Collection interval
- Writable flag
- Unit

Supports CSV/XLSX batch import/export of data point configurations.

9.7 Modbus Register Mapping

BACnet object types are automatically mapped to corresponding Modbus register types:

- AI/MSI → Input Register (FC 04, read-only)

- AO/AV/MSO/MSV → Holding Register (FC 03, read/write)
- BI → Discrete Input (FC 02, read-only)
- BO/BV → Coil (FC 01, read/write)

Click "Export Modbus Map" to export the register mapping table as a CSV file.

9.8 Real-time Monitoring & Control

Click the "Control Panel" button to open the real-time monitoring dashboard: real-time value display, manual write (triggering BACnet WriteProperty), and communication log.

9.9 Breakpoint Resume

When breakpoint resume is enabled, if the northbound connection is interrupted, pending data is automatically cached to local files and retransmitted when the connection is restored.

9.10 Troubleshooting

- Service fails to start: Verify configuration is correct, check system logs
- BACnet device connection failure: Confirm the target IP is reachable and port 47808 is not blocked by the firewall
- Object read returns error: Verify that the BACnet object type and instance number match the DDC configuration
- Modbus TCP not responding: Confirm the slave ID is configured correctly and port 502 is not occupied
- MQTT connection failure: Check the Broker address, port, and username/password

Chapter 10 Energy Management Service

The Energy Management Service is an energy consumption monitoring and management module on the TR400, providing device management, energy analysis, carbon management, operations management, and system settings capabilities, suitable for industrial and commercial energy management scenarios.

10.1 Dashboard

The Dashboard page provides a global overview of energy management, including key metrics such as total power, today's energy consumption, today's water usage, active alerts, and online devices. The service status is displayed at the top, with a "Start Service" button to enable the energy management service.

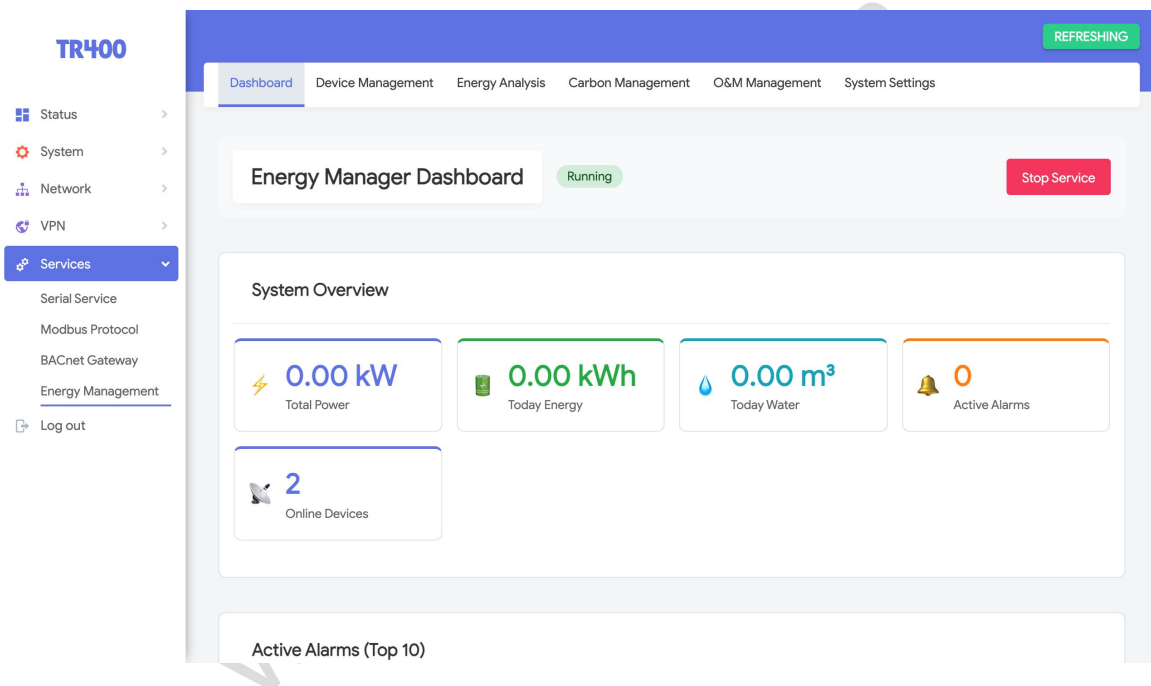


Figure 10-1: Energy Management Dashboard

10.2 Device Management

The Device Management page is used to add and manage energy monitoring devices. It supports device list viewing, device addition, product definition, variable classification, and control management.

- **Device List:** View all configured energy monitoring devices, including device name, energy type, protocol type, host/port, enabled status, and online status
- **Product Definition:** Define device product models and data point templates
- **Variable Classification:** Manage data collection variables and categories
- **Control Management:** Configure remote control parameters for devices

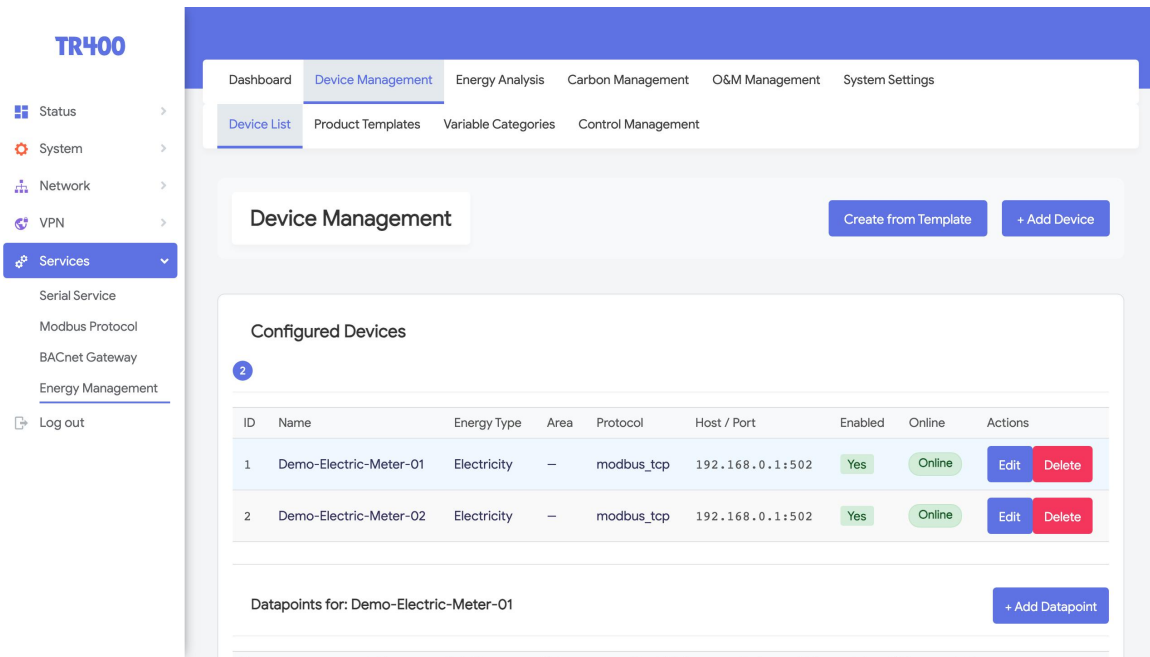


Figure 10-2: Device Management - Device List

10.3 Energy Analysis

The Energy Analysis page provides multi-dimensional energy consumption data viewing and analysis features, including the following sub-functions:

- **Real-time Data:** Auto-refreshes every 3 seconds, displaying real-time collected values from all devices, including voltage, current, power, and energy parameters
- **Historical Data:** Provides trend charts and tabular views of energy consumption data, with filtering by time range, device, and data type
- **Time-of-Use Pricing:** Configure peak/valley/flat rate time periods and view consumption and cost distribution by period
- **Energy Reports:** Generate energy consumption analysis reports, supporting multiple time ranges and data export formats

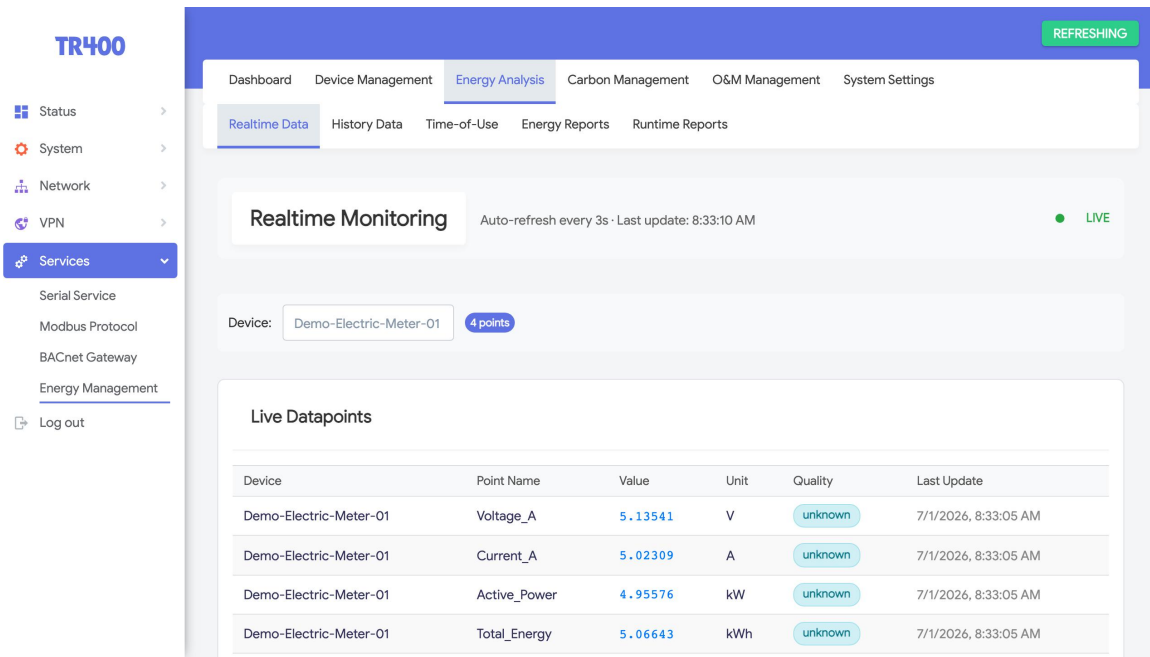


Figure 10-4: Energy Analysis - Real-time Data

10.4 Carbon Management

The Carbon Management page provides carbon emission monitoring and calculation features, including the following sub-functions:

- **Emission Factors:** Configure carbon emission conversion factors for various energy types
- **Carbon Footprint:** Calculate carbon emissions based on energy consumption data and track carbon footprint trends
- **Carbon Ledger:** Record carbon emission details, supporting export and auditing
- **Carbon Calculation:** Customize carbon emission calculation rules and parameters

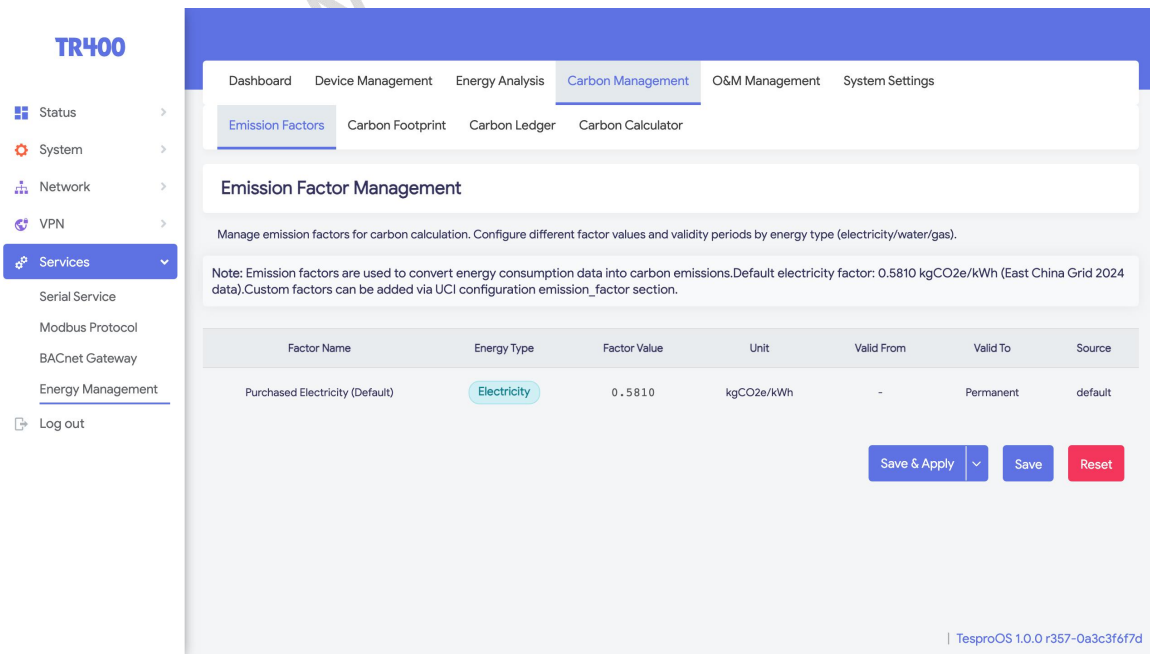


Figure 10-6: Carbon Management - Carbon Calculation

10.5 Operations Management

The Operations Management page provides device operations and alarm management features, including the following sub-functions:

- **Alarm Center:** Real-time monitoring of alarm status, displaying statistics for active, critical, warning, and informational alarms, supporting both real-time and historical alarm viewing
- **Alarm Rules:** Define alarm conditions for energy consumption anomalies, including threshold alarms, rate-of-change alarms, and offline alarms, with configurable trigger conditions, notification methods, and severity levels
- **Notification Configuration:** Configure alarm notification delivery methods, including email, SMS, and WebHook channels
- **Event Log:** Record and view system operation events and operation logs
- **Pricing Configuration:** Set electricity billing rules, including peak/valley/flat rate time periods and corresponding rates for automatic electricity cost calculation

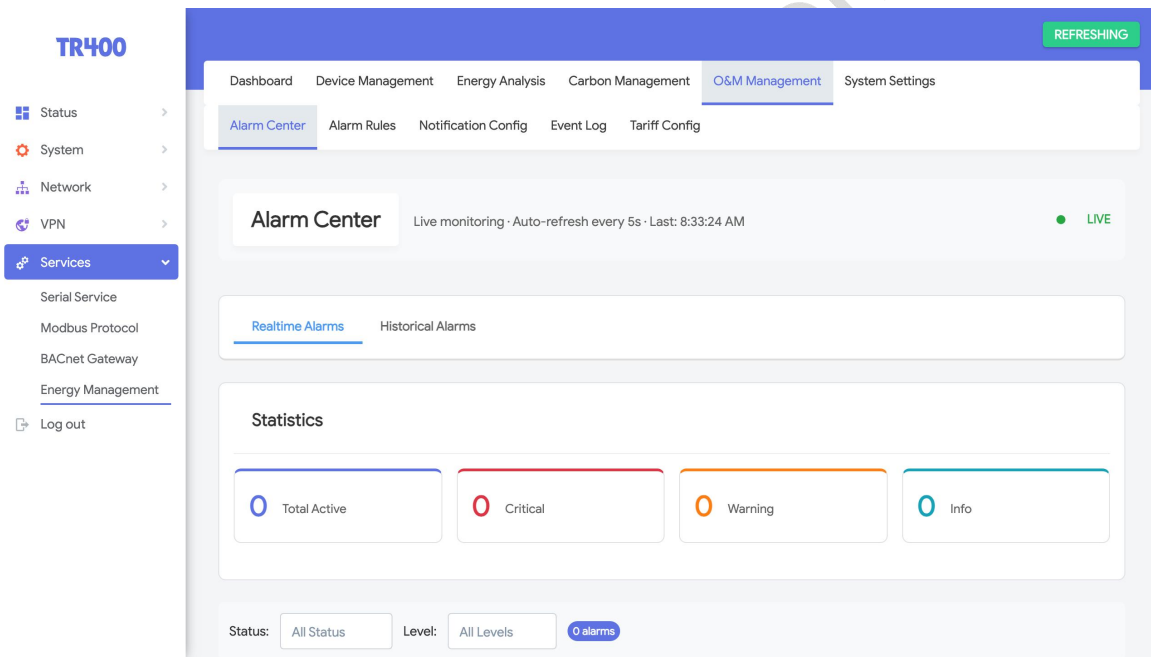


Figure 10-8: Operations Management - Alarm Center

10.6 System Settings

The System Settings page configures basic parameters of the Energy Management Service, including the following functions:

- **Service Control:** Start, stop, and restart the Energy Management Service, with auto-start on boot option
- **Data Simulator:** Generate simulated meter data for testing when no physical devices are connected
- **General Settings:** Configure deployment mode (flash storage), data root path, and collection thread count

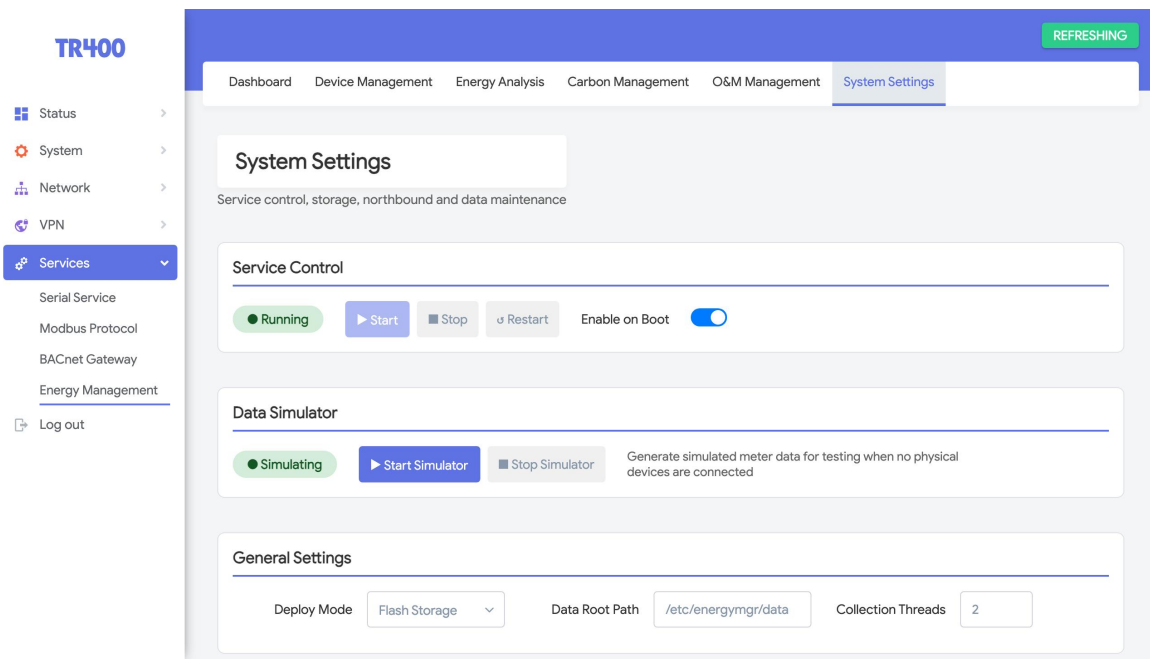


Figure 10-10: System Settings

Chapter 11 Cloud Device Management Service

The TR400 supports remote management via the Tespro Cloud Device Management Platform (mgr.tespro.com). This platform provides device status monitoring, remote Web access, package distribution, and online firmware upgrade capabilities.

11.1 Service Overview

The Tespro Device Manager is a cloud-based industrial router management platform with the following key features:

- Real-time device status monitoring: Receive device operational status data via the MQTT protocol
- Remote Web access: Access the device Web management interface remotely via WebSocket tunnels
- Package management: Distribute and install software packages on devices
- Online firmware upgrade: Centrally manage and push firmware updates
- Device group management: Organize devices into groups for batch operations

11.2 Registration & Login

Visit <https://mgr.tespro.com> to access the device management platform.

New users need to register: Enter your email address, send a verification code, and complete registration.

Registered users log in: Enter your email, obtain a login verification code, verify, and log in.

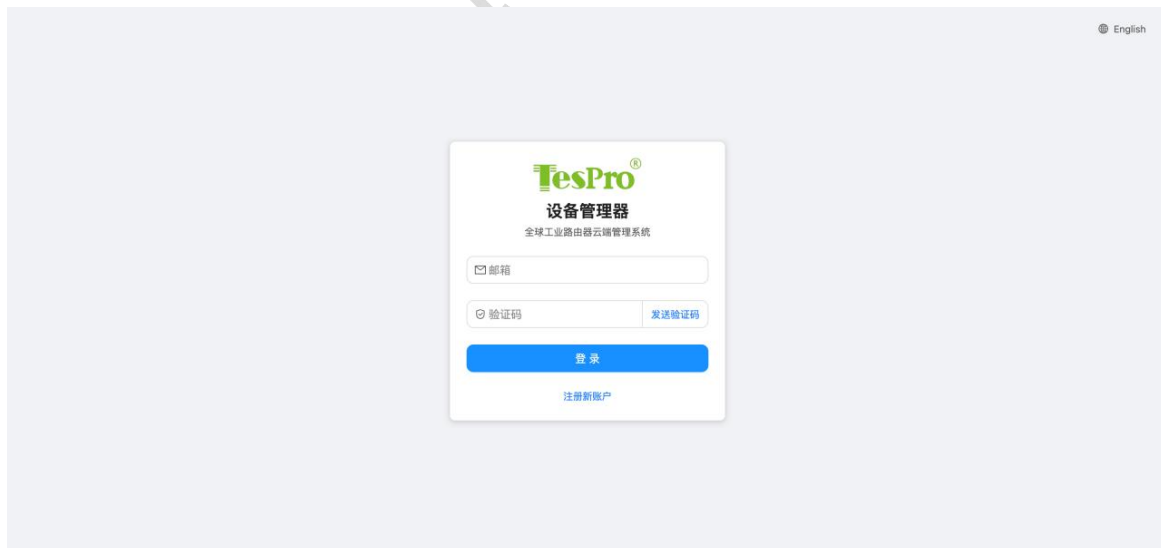


Figure 11-1: Tespro Device Manager Login Page

11.3 Remote Device Access

When the device's remote access tunnel is enabled and successfully connected to the cloud platform, you can access the device remotely via its Web access URL.

Remote access URL format: `https://<subdomain>.mgr.tespro.com`

Tip: The remote access URL can be found on the device Web interface at **System > Device Online Manager > Settings**.

11.4 Connection Architecture

The communication between the TR400 and the Cloud Management Platform uses the following architecture:

This architecture enables devices to be remotely accessed and managed without a public IP address, making it particularly suitable for devices deployed behind NAT or firewalls.

- Status reporting channel: The device periodically reports operational status to the cloud platform via the MQTT protocol (TLS encrypted, port 8883)
- Remote access channel: A bidirectional communication channel between the device and cloud platform is established via WebSocket tunnels (TLS encrypted, port 443)
- All communications support TLS encryption to ensure data transmission security

Appendix A: Technical Specifications

A.1 Hardware Specifications

Parameter	Specification
Processor	Dual-core ARM Cortex-A53 @ 1.3GHz
Memory	512MB
Storage	256MB Flash
Ethernet	5× Gigabit Ethernet ports (4 LAN + 1 WAN)
WiFi	WiFi 6 AX3000: 2.4GHz 574Mbps + 5GHz 2402Mbps
Serial Ports	2× RS485 + 1× RS232, UDP multicast support
Expansion Slot	1× Mini PCIe (4G/5G module)
Power	DC 12V/2A
Dimensions	150×115×35mm
Operating Temperature	-40°C ~ 70°C
Protection Rating	IP30

A.2 Software Specifications

Parameter	Specification
Operating System	TesproOS
Firmware Version	TesproOS 1.0.0
Firewall	nftables (firewall4)
VPN	OpenVPN / WireGuard / IPsec / PPTP
Modbus	RTU/TCP Master + Slave dual role, supports 64 devices and 128 data points

BACnet	IP/MS/TP data collection, 64 devices and 128 data points
Energy Management	12 sub-function modules
Northbound Protocols	MQTT / TCP / UDP / HTTP / OPC UA Client / OPC UA Server

Appendix B: Default Configuration

Parameter	Default Value
LAN IP	192.168.0.1
LAN Subnet	192.168.0.0/24
DHCP Range	192.168.0.100 - 192.168.0.200
WiFi SSID (2.4G)	TR400-2.4G (password: 12345678)
WiFi SSID (5G)	TR400-5G (password: 12345678)
WiFi Password	12345678
Admin Username	root
Admin Password	000000
SSH Port	22
HTTP Port	80
NTP Server	pool.ntp.org

Appendix C: Troubleshooting

Unable to Access the Web Management Interface

- Verify that the Ethernet cable is securely connected to a LAN port (not the WAN port).
- Ensure your computer is configured to obtain an IP address automatically (DHCP).
- Try clearing the browser cache or using a different browser.
- Try accessing <http://192.168.0.1>.

WiFi Not Working

- Confirm that WiFi is enabled under Network > WiFi Network.
- Ensure the antenna is properly connected.
- Try switching to a different WiFi channel to avoid interference.

No Internet Connection

- Verify that the WAN port is connected to your modem or upstream network.
- Check the WAN interface status under Network > Network Interfaces.
- Use the diagnostic tool to Ping an external IP (e.g., 8.8.8.8).

4G/5G Cellular Network Not Connecting

- Confirm that the SIM card is properly inserted and the module is recognized.

- Check signal strength under Status > Cellular Network.
- Verify that the APN settings match your carrier's requirements.

Modbus Communication Error

- Verify that the serial parameters (baud rate, parity, data bits, stop bits) exactly match those of the slave device.
- Check physical wiring: RS485 requires correct A/B polarity and proper termination resistors.
- Confirm that the slave address is correct and unique on the bus.
- Increase the timeout value if devices respond slowly.
- Use the Serial Service's Debug & Test tool to verify raw serial communication first.

VPN Connection Issues

- Verify that the VPN package is properly installed.
- Check the server address, port, and authentication credentials in the VPN configuration.
- Review system logs for connection errors.
- Ensure the firewall is not blocking the ports and protocols required by the VPN.

Forgot Administrator Password

If you have forgotten the administrator password, perform a factory reset on the router:

1. Power on the router and wait for it to fully boot (approximately 60 seconds).
2. Press and hold the reset button for more than 10 seconds until the LED blinks rapidly.
3. Release the button. The router will restore factory settings and restart.
4. Connect via Ethernet and access <http://192.168.0.1>, then log in with username root and password 000000.

Warning: A factory reset will erase all custom configurations. Create regular backups via System > Backup.

Contact Us

Thank you for choosing the TR400 Industrial Router. If you have any questions or requirements, please do not hesitate to contact us.

Tespro Electronics Co., Ltd.

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